
C²U in RAG: Compositional Concept Unlearning in Retrieval-Augmented Generation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Retrieval-Augmented Generation (RAG) improves generation by grounding re-
2 sponses in retrieved documents, but it also creates an exposure pathway: unwanted
3 or sensitive evidence may be retrieved and passed to the generator. Existing
4 approaches often require retraining, re-encoding, or rebuilding indexes, while
5 lightweight filtering and reranking provide limited concept-level controllability.
6 We introduce *Compositional Concept Unlearning* (C²U), a retrieval-side framework
7 that suppresses target concepts through inference-time query editing, without updat-
8 ing the retriever backbone or rebuilding the index. C²U learns a sparse atom-space
9 representation for concept-profile estimation and targeted atom steering, enabling
10 localized suppression of concept-aligned query components. Across five datasets
11 and adversarial query settings, C²U provides a favorable suppression–utility trade-
12 off compared with strong erasure and editing baselines, reducing target retrieval
13 and token-budgeted context exposure while preserving non-target retrieval utility.
14 We further analyze dictionary size, profile sensitivity, multi-concept suppression,
15 backbone transfer, and cross-domain adaptation, showing that the atom dictionary
16 can be extended with lightweight incremental updates when domain shift occurs.

17 1 Introduction

18 Retrieval-Augmented Generation (RAG) is widely used today. However, its reliance on external
19 documents introduces safety and governance risks, including private information leakage, biased
20 or incorrect content, and liability concerns. To address these challenges, unlearning techniques
21 selectively remove or suppress undesired content from RAG systems while preserving utility. This
22 leads to a critical research question:

23 *How can we achieve precise and efficient unlearning in RAG while preserving model utility?*

24 Current approaches to RAG safety predominantly rely on output-side intervention, often referred
25 to as “guardrails” or “filters” [13]. These methods employ auxiliary models to detect and suppress
26 sensitive content during generation or retrieval. While easy to deploy, these guardrails have no direct
27 control over the retrieval operation. Specifically, if concept-bearing evidence is retrieved into context,
28 adversarial prompting can still trigger exposure. Recent studies on adversarial attacks demonstrate
29 that as long as the underlying knowledge persists in the model’s representations, safety mechanisms
30 can be easily circumvented via “jailbreaking” techniques, such as adversarial suffixes [30] or latent
31 instruction embeddings [20].

32 This motivates retrieval-side intervention: if target-bearing evidence enters the context, downstream
33 guardrails or generators may still expose it under adversarial prompting. Suppressing target evidence
34 before context construction is therefore a necessary and practical point of control. However, existing
35 approaches face a deployment trade-off: retraining or model-editing methods may require encoder

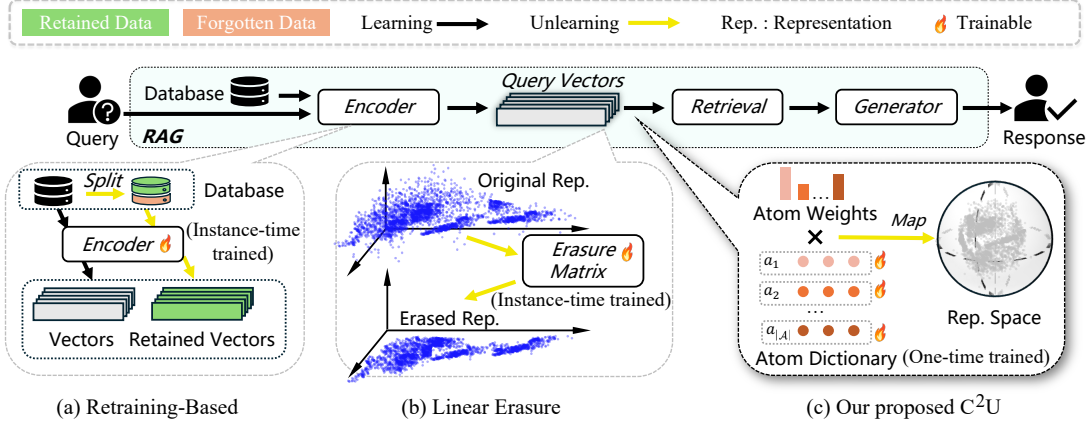


Figure 1: **Three Unlearning Paradigms for RAG.** (a) **Retraining/Editing**: updates the encoder/index using retained and forgotten data. (b) **Linear Erasure**: removes concepts via a global erasure transform in representation space. (c) **C²U (Ours)**: steers only query vectors at retrieval time using an atom dictionary and sparse weights, without retriever retraining or index rebuilding.

updates or index refreshes, while linear erasure applies a global transform that can be too coarse for concepts distributed across multiple semantic factors.

In response, we propose *Compositional Concept Unlearning (C²U)* in RAG systems as shown in Figure 1(c). Unlike retraining or single-direction erasure, atom-mixture editing localizes suppression to a small set of activated factors, reducing collateral damage to non-target semantics. C²U operationalizes compositionality as sparse and reusable control over retrieval representations. Rather than applying a single global erasure direction, it learns an atom-space representation in which concept profiles can be estimated and target-aligned query components can be steered locally. The learned atoms serve as retrieval-relevant factors that support coherent profiles, localized editing, and predictable suppression–utility behavior.

C²U employs an *Atomic Compositional Representation (ACR)* module to project query embeddings into this atom space. After one-time ACR training, unlearning is performed purely at inference time by editing query-side atom weights, without updating the retriever backbone or rebuilding the document index.

We evaluate C²U on five datasets under benign and adversarial query settings, and compare against linear erasure (INLP [21], LEACE [4]) and model editing (DARE-GRL[10, 27], PISCES [11]) baselines. Across settings, C²U reduces target retrieval and context exposure, including token-budgeted exposure under attacks, while preserving non-target retrieval utility in the regimes where aggressive perturbation baselines tend to over-edit and linear erasure baselines tend to under-suppress.

In summary, we introduce a drop-in concept unlearning mechanism for RAG and show it improves the suppression–utility trade-off and robustness under attacks.

Our main contributions are as follows:

- **Retrieval-side unlearning formulation.** We target the retrieval-side exposure pathway in RAG, suppressing target-bearing evidence before it enters the context while preserving non-target retrieval utility.
- **Compositional query editing.** We introduce C²U, an inference-time query editing framework that learns a sparse atom-space representation for concept-profile estimation and localized atom steering, without retriever-backbone retraining or index rebuilding.
- **Robust suppression and utility analysis.** We evaluate C²U across five datasets under standard and adversarial queries, showing favorable suppression–utility trade-offs and reduced token-budgeted context exposure.
- **Operating-regime analysis.** We provide ablations on dictionary granularity, concept-profile robustness, simultaneous multi-concept suppression, backbone transfer, entity-centric behavior, domain-shift adaptation, and deployment cost.

70 2 Related Work

71 2.1 Editing in Retrieval-Augmented Generation

72 RAG enhances LLMs with external documents, improving factual grounding and domain coverage.
73 A large body of work [12, 2, 28, 26, 15] improves retrieval quality and efficiency by filtering or re-
74 ranking mechanisms. However, recent studies highlight that RAG is not inherently safer. Comparative
75 evaluations demonstrate that unsafe or private content can surface even when both the model and
76 the retrieved documents appear to be safe [1], and that knowledge poisoning attacks on the retrieval
77 source can subvert downstream generation [29]. Frameworks such as SafetyRAG [14] aim to mitigate
78 these risks through external moderation and guardrails, but they typically operate at the document or
79 string level and require manual rule updates when policies change, lacking the ability to suppress
80 specific concepts within documents. Moreover, such approaches are less suited to concept-level
81 control that directly targets retrieval exposure in embedding-based pipelines.

82 Unlearning in RAG has been explored via coarse database redaction or constrained optimization at
83 the knowledge-base level [23]. While effective for removing entire entries, such approaches lack the
84 ability to act at concept granularity or to support zero-shot suppression of specific entities or themes
85 without retraining. In contrast, our setting emphasizes lightweight, inference-time suppression of
86 target concepts without modifying the underlying index, making it practical for rapidly evolving
87 policies.

88 2.2 Machine Unlearning and Conceptualization

89 **Machine unlearning** focuses on selective knowledge editing and undesirable content filtering.
90 Several unlearning methods [6, 22, 8] have been proposed to modify or remove specific factual
91 knowledge. However, concept-agnostic approaches often suffer from related concept ambiguity
92 and partial semantic removal, leading interventions to spill over into semantically adjacent content.
93 Representation-level concept removal methods (e.g., post-hoc linear erasure on frozen embeddings
94 such as INLP [21] and LEACE [4]) offer lightweight baselines but can be limited in handling
95 compositional concepts and fine-grained exposure control.

96 **Concept erasure** has emerged as a more comprehensive approach to overcome those limitations
97 in diffusion models [19, 9]. EraseDiff [25] formulates a constrained optimization problem using
98 value functions to reduce complexity. Adaptive Guided Erasure (AGE) [5] demonstrates that concept
99 spaces exhibit geometric and locality characteristics, such that erasure affects semantically relevant
100 concepts. While promising, these methods are typically studied in parameter-updating settings,
101 whereas retrieval systems often impose deployment constraints that require the index to remain fixed,
102 and interventions must be lightweight at inference time.

103 Our work addresses the critical gap between concept-driven methods and knowledge editing tech-
104 niques in RAG by focusing on inference-time, concept-level suppression for retrieval. Rather than
105 relying on corpus redaction or retriever retraining, C²U provides fine-grained control over retrieval
106 exposure by targeting suppression of unwanted concepts, while preserving semantic relevance for
107 non-target queries.

108 3 Background

109 3.1 Retrieval-Augmented Generation

110 RAG combines a retriever R_θ and a generator ϕ to support knowledge-intensive generation by
111 grounding outputs in an external corpus $\mathcal{D}$. Let $E_\theta(\cdot)$ denote the embedding function, and $s(q, d) =$
112 $\langle E_\theta(q), E_\theta(d) \rangle$ the similarity score. Given a query q , the retriever ranks documents by $s(q, d)$ and
113 returns a top- k set $\mathcal{D}_q \subset \mathcal{D}$.

114 3.2 Retrieval-Stage Unlearning in RAG

115 A common formulation of retrieval-stage unlearning in prior work is training-based: the corpus is
116 partitioned into a retain subset $\mathcal{D}_R$, and a forget subset $\mathcal{D}_F$. The retriever is optimized to preserve
117 retrieval on $\mathcal{D}_R$ while suppressing retrieval of $\mathcal{D}_F$ via contrastive objectives [23]. However, training-

x : Input texts q : Query $\mathcal{D}_c$: Concept-related texts a_i : Atom i 🔥 : Trainable ➡ : Training Process
 w_x : Atom Weights π_c : Concept Profile z_x : Reconstructed Rep. ❄️ : Frozen ➡ : Inference Process

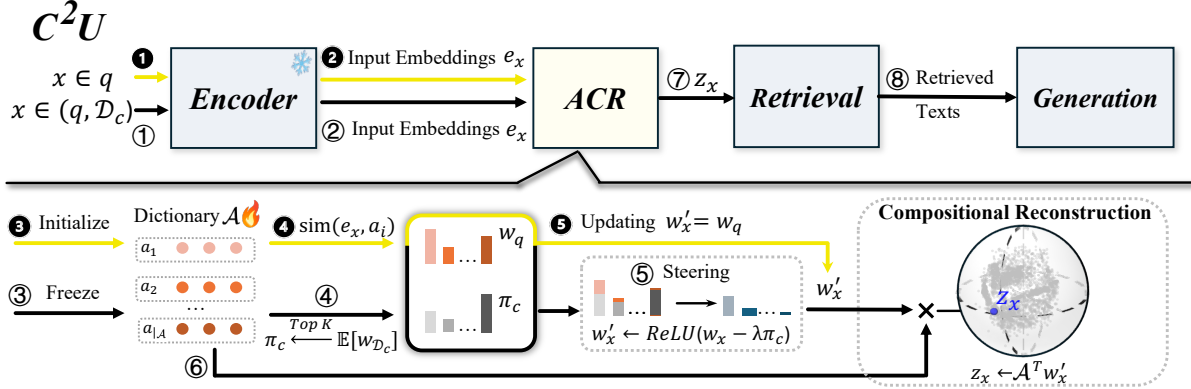


Figure 2: Overview of the C²U Framework. An input text is encoded into a **peaked** atom-weight distribution over a latent atomic dictionary and composed into a cosine-retrieval-compatible embedding that supports concept-level diagnosis and suppression.

based unlearning can be operationally heavy and is often less structured for fine-grained, concept-level control.

In this work, we focus on a different practical setting: the corpus index remains fixed, and policies are enforced by lightweight, inference-time edits on retrieval representations to reduce *retrieval exposure* of target concepts.

4 Methodology

4.1 The C²U Framework: An Overview

Figure 2 presents an end-to-end view of C²U. We retain the standard RAG modules and insert the Atomic Compositional Representation (ACR) module between the encoder and the retrieval index. At inference time, given a target concept, C²U estimates an atom-space concept profile, edits only the query-side atom weights, and recomposes the edited query embedding for retrieval. The retriever backbone, document index, and generator remain fixed. Thus, policy updates are implemented as query-side steering rather than retriever retraining or index rebuilding.

4.2 Atomic Compositional Representation

The ACR module converts an input text into an atom-space representation that can be recomposed into a dense retrieval embedding. Its goal is to preserve the retrieval geometry of the frozen encoder while exposing sparse atom weights that can be edited for concept-level steering.

ACR consists of a learnable atom dictionary, a sparse weight generator, and a compositional reconstruction module. Given a frozen encoder representation, the weight generator assigns a peaked distribution over atoms, and the reconstruction module composes the selected atoms into a normalized dense embedding compatible with cosine retrieval.

4.2.1 Atom Dictionary

The AD $\mathcal{A} = \{a_1, \dots, a_{|\mathcal{A}|}\}$ consists of N learnable atom vectors, where each $a_i \in \mathbb{R}^d$. The atoms carry no pre-assigned semantics. Instead, they are learned from retrieval supervision jointly with a lightweight projection head atop a frozen encoder backbone. Intuitively, $\mathcal{A}$ serves as a shared compositional basis that is reused across queries and documents.

During training, the similarity-based scoring and temperature-scaled softmax in SWG provide a differentiable path to update the atom dictionary. After training, the atom dictionary is fixed during

inference. Concept profiles are constructed from the atom weights assigned by SWG, rather than from manually labeled atom semantics.

We initialize $\mathcal{A}$ by randomly sampling from a standard normal distribution and rely on the retrieval objective in Equation (3) to shape the atoms into sparse, reusable factors that capture recurring retrieval-relevant structure.

4.2.2 Sparse Weights Generation

We implement the atom–input similarity as a bilinear score between the frozen encoder representation and each atom, using a lightweight trainable projection shared across atoms.

Given an input text x , we first obtain an encoder representation $h_x \in \mathbb{R}^d$ from a frozen backbone. The SWG then assigns a relevance weight to each atom in the learned dictionary $\mathcal{A}$ by scoring atom–input compatibility via a lightweight projection head. Concretely, each atom $a_i \in \mathcal{A}$ receives a score through a learnable similarity function $\text{sim}(h_x, a_i)$, and we apply a temperature-scaled softmax to obtain approximately sparse atom weights:

$$w_{x,i} = \frac{\exp(\text{sim}(h_x, a_i) / \tau)}{\sum_{j=1}^{|\mathcal{A}|} \exp(\text{sim}(h_x, a_j) / \tau)}, \quad (1)$$

where $|\mathcal{A}|$ is the number of atoms, $\tau > 0$ controls distribution sharpness (smaller τ yields more peaked mixtures), and $\text{sim}(\cdot, \cdot)$ is implemented by a trainable projection layer on top of the frozen encoder. The resulting weight vector $w_x = (w_{x,1}, \dots, w_{x,|\mathcal{A}|})$ serves as atom weights describing the input as a mixture over latent atoms. These weights are later interpreted and edited for concept-level suppression. To facilitate precise edits, we additionally encourage peaked mixtures by penalizing the entropy $H(w_x)$ during training (see implementation details).

4.2.3 Compositional Reconstruction

The CR module aggregates the atom weights w_x over the atom dictionary $\mathcal{A}$ to form a dense embedding used for retrieval. Concretely, let $\mathcal{A} \in \mathbb{R}^{|\mathcal{A}| \times d}$ denote the atom matrix whose rows are atoms a_i . Given the weight vector $w_x \in \mathbb{R}^{|\mathcal{A}|}$, we compose the embedding by an atom-weighted sum and apply ℓ_2 normalization:

$$z_x = \frac{\mathcal{A}^\top w_x}{\|\mathcal{A}^\top w_x\|_2} = \frac{\sum_{i=1}^{|\mathcal{A}|} w_{x,i} a_i}{\left\| \sum_{i=1}^{|\mathcal{A}|} w_{x,i} a_i \right\|_2} \in \mathbb{R}^d. \quad (2)$$

This normalized embedding is directly compatible with cosine-similarity-based retrieval. This constructive path, from atoms to weights, then to a normalized dense embedding, enables concept-level unlearning through subsequent manipulation of the atomic mixture w_x .

4.3 Training Paradigm

We train ACR to learn an atomic dictionary that preserves the semantic structure of a pre-trained dense retriever while enabling controllable composition. The same encoder is applied to both queries and documents. During training, we freeze the transformer backbone and optimize only the atom dictionary $\mathcal{A}$ and a lightweight projection/composition head.

Given a query q , a positive document d^+ , and a negative document d^- , we obtain their composed embeddings z_q, z_{d^+}, z_{d^-} and optimize a margin-based triplet objective in cosine space:

$$\mathcal{L}_{\text{ret}} = \max(0, m_r - z_q^\top z_{d^+} + z_q^\top z_{d^-}), \quad (3)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity and $m_r > 0$ is a margin. Since z_x is ℓ_2 -normalized, $z_q^\top z_d$ equals cosine similarity. To preserve utility, we add a cosine alignment term that encourages each composed embedding to remain close to the corresponding frozen retriever embedding. We further include an entropy penalty on the atom-weight distribution to encourage peaked mixtures, which facilitates precise edits in subsequent unlearning (full objective in Section A.3).

186 **From training to inference-time editing.** After ACR training, we freeze all parameters (E , P , and
 187 $\mathcal{A}$) and construct the document index once using ACR-composed document embeddings. Subsequent
 188 concept-unlearning updates do not require re-encoding documents or rebuilding the index; only
 189 query-side atom weights are edited at inference time. At inference, we keep document embeddings
 190 fixed and apply a parameter-free wrapper that edits only the query-side atom weights w_q into w'_q ,
 191 then recomposes z'_q for retrieval with the same scoring function. The atom dictionary is fixed for
 192 inference-time unlearning within a deployment domain. Appendix D.9 studies cross-domain transfer
 193 and an optional lightweight dictionary-extension ablation for domain shifts.

194 4.4 Concept-Level Unlearning

195 We implement Steps ⑤ and ⑥ in C²U (Figure 2) as a two-stage, parameter-free procedure on top of
 196 the trained ACR (Section 4.2). Given an input x , ACR produces atom weights $w_x \in \mathbb{R}^{|\mathcal{A}|}$ over atoms
 197 and a corresponding representation z_x . Concept-level unlearning is achieved by (i) constructing an
 198 atom-space profile for a target concept; and (ii) steering w_x along this profile to obtain an edited
 199 representation z'_x that downweights the concept while preserving unrelated semantics.

200 4.4.1 Atomic Concept Profiles

201 For a target concept c , we construct an atom-space profile from a small set of associated documents.
 202 Unless otherwise specified, we use 20 profile documents and evaluate profile-size and contamination
 203 sensitivity in Appendix D.3. Let $\mathcal{D}_c$ denote the profile set. Running ACR on each $x \in \mathcal{D}_c$ yields atom
 204 weights $w_x \in \mathbb{R}^{|\mathcal{A}|}$. We compute the mean atom-weight vector and apply a non-negativity filter:

$$\tilde{\pi}_c = \text{ReLU}(\mathbb{E}_{x \in \mathcal{D}_c}[w_x]), \quad \pi_c = \frac{\tilde{\pi}_c}{\sum_{i=1}^{|\mathcal{A}|} \tilde{\pi}_c[i]}. \quad (4)$$

205 We then retain only the top- K_{atom} coordinates of π_c , set the rest to zero, and re-normalize. This yields
 206 a compact concept profile used for inference-time steering. We tune K_{atom} on a validation split and
 207 use $K_{\text{atom}} = 30$ by default for the 1024-atom dictionary, with sensitivity analysis in Appendix D.2.

208 4.4.2 Atom-Space Steering

209 To suppress concept c at inference time, we directly edit the atom weights w_x using the concept
 210 profile π_c . We first compute an adaptive strength based on cosine alignment:

$$\alpha_x = \max(0, \cos(\hat{w}_x, \hat{\pi}_c)), \quad (5)$$

211 where $\hat{w}_x$ and $\hat{\pi}_c$ denote ℓ_2 -normalized vectors. We apply $\max(0, \cdot)$ to ensure suppression is activated
 212 only when the query is positively aligned with the target concept, avoiding unnecessary edits on
 213 unrelated queries. Given a suppression coefficient $\lambda > 0$, we set $\gamma_x = \lambda \cdot \alpha_x$ and apply profile-guided
 214 subtraction followed by a non-negativity filter and ℓ_1 re-normalization:

$$\tilde{w}_x = \text{ReLU}(w_x - \gamma_x \pi_c), \quad w'_x = \frac{\tilde{w}_x}{\sum_{i=1}^{|\mathcal{A}|} \tilde{w}_x[i]}. \quad (6)$$

215 We then reconstruct the edited dense embedding via the same atom composition:

$$z'_x = \frac{\sum_{i=1}^{|\mathcal{A}|} w'_{x,i} a_i}{\left\| \sum_{i=1}^{|\mathcal{A}|} w'_{x,i} a_i \right\|_2}. \quad (7)$$

216 This procedure is fully non-parametric, with no model parameters updated, and only the atomic
 217 mixture is edited and recomposed.

218 5 Experiments

219 5.1 Experimental Setup

220 **Datasets and Metrics.** We evaluate on five diverse benchmarks: ArXiv, TOFU, Amazon, IMDB, and
 221 MNLI [16, 17, 3, 24].¹ Each dataset defines *target* concepts (to unlearn) and *control* concepts (to

¹arXiv dataset (Kaggle), <https://www.kaggle.com/dsv/7548853>, accessed on Jan. 28, 2026.

Table 1: **RAG retrieval unlearning results.** Higher Target Cos. Drop and Hit@K Drop indicate stronger suppression; lower Control/Neighbor Delta and Target Exposure Delta indicate better utility preservation and safety.

Dataset	Method	Target Cos.	Hit@K	Ctrl./Neigh. Delta	Target Exp. Delta
Amazon	C ² U	0.276	<u>0.500</u>	<u>-0.039</u>	<u>-0.240</u>
	INLP	0.264	0.270	-0.019	0.088
	LEACE	0.000	0.260	-0.017	-0.052
	PISCES	0.001	0.100	-0.007	-0.024
	DARE	0.806	0.580	-0.042	-0.344
	Random	0.811	0.200	0.014	0.212
ArXiv	C ² U	0.030	0.286	0.024	-0.098
	INLP	0.031	0.058	0.019	0.080
	LEACE	0.007	0.062	<u>0.002</u>	<u>-0.042</u>
	PISCES	0.001	0.002	0.001	0.008
	DARE	<u>0.885</u>	<u>0.790</u>	0.866	0.091
	Random	0.926	0.824	0.871	0.142
IMDB	C ² U	-0.481	1.000	<u>0.032</u>	<u>-0.952</u>
	INLP	0.060	0.880	0.636	-0.716
	LEACE	-0.258	1.000	0.158	-0.968
	PISCES	-0.039	0.000	0.000	0.032
	DARE	0.485	0.000	0.244	0.032
	Random	<u>0.476</u>	0.670	0.300	-0.526
MNLI	C ² U	0.146	0.610	0.098	<u>-0.142</u>
	INLP	0.041	0.090	0.028	0.164
	LEACE	0.010	0.230	0.000	-0.060
	PISCES	0.000	0.000	<u>0.000</u>	0.014
	DARE	0.898	0.900	0.453	-0.150
	Random	<u>0.832</u>	<u>0.650</u>	0.576	0.270
TOFU	C ² U	0.114	0.350	0.068	-0.014
	INLP	<u>0.863</u>	0.850	0.645	-0.038
	LEACE	0.001	<u>0.800</u>	<u>0.127</u>	<u>-0.036</u>
	PISCES	0.002	0.000	0.147	0.198
	DARE	0.862	0.850	0.653	0.962
	Random	0.981	0.850	0.659	0.008

Note: Bold and underline denote best and second-best results. Values are rounded to 3 decimals; entries shown as 0.000 may denote magnitudes below 0.001. DARE abbreviates DARE-GRL.

222 preserve). We use fixed-index dense retrieval and report **Hit@K**, **Exposure@K**, and **Target Cos**
223 **Drop** for suppression, alongside **Control Delta** for utility. We distinguish between profile truncation
224 K_{atom} and retrieval cutoff K_{eval} . Detailed statistics are in Appendix B. Formal metric definitions are
225 in Appendix A.

226 **Baselines.** We compare C²U against: (i) **Linear Erasure:** INLP [21] and LEACE [4]; (ii) **Model**
227 **Editing:** PISCES [11] and DARE-GRL [27]; and (iii) **Random:** Gaussian noise Injection. Unlike
228 parameter-editing baselines, our method and linear erasure operate at inference time without re-
229 indexing. Implementation details are in Appendix C.

230 **Implementation and Threat Model.** We tune hyperparameters (K_{atom} , λ) on a validation split (see
231 Table 3). We evaluate robustness under a black-box threat model via three query variants: Standard
232 direct queries, Injection prompts, and Zou-noise adversarial suffixes [30]. Training protocol
233 and robustness metrics are detailed in Appendix A. Unless otherwise specified, all main results use
234 all-MiniLM-L6-v2 as the default backbone for controlled comparison. We additionally evaluate
235 gte-base, bge-base-en-v1.5, and e5-base-v2 in Appendix D.8.

236 5.2 Experimental Results

237 5.2.1 Exp-I: Unlearning Efficacy and Safety under Standard Retrieval

238 Table 1 compares C²U with baselines under standard retrieval. Effective retrieval unlearning requires
239 both target suppression, measured by Hit@K Drop, and non-target preservation, measured by
240 Control/Neighbor Delta.

Linear erasure methods are competitive when the target concept is concentrated in a low-dimensional direction, as seen on IMDB and Amazon, but are less effective on distributed concepts such as ArXiv and MNLI. In contrast, DARE-GRL and Random can produce large hit reductions, but often by broadly disrupting the retrieval neighborhood, leading to large Control/Neighbor Delta. PISCES has limited impact in this retrieval setting, consistent with the mismatch between its original generative-editing objective and dense retrieval unlearning.

C^2U provides a more favorable suppression–utility trade-off in settings where linear methods under-suppress distributed concepts. On ArXiv and MNLI, it achieves meaningful Hit@K Drop while preserving the non-target neighborhood substantially better than aggressive perturbation baselines. TOFU is more challenging because target and non-target facts are tightly coupled around author identifiers. Under the conservative default setting in Table 1, C^2U sacrifices some hit suppression to better preserve adjacent non-target retrieval. Appendix D.6 reports a separate per-query diagnostic showing that stronger target suppression is possible, but mainly at the cost of non-target spillover.

One exception is IMDB, where Target Cos Drop becomes negative despite perfect Hit@K suppression. We attribute this to multi-modal target clusters, where centroid-based proxies are less reliable than ranking- and exposure-based metrics. Overall, Exp-I shows that hit suppression alone is insufficient: C^2U often better aligns target suppression with utility preservation, while linear methods remain competitive for lower-dimensional or tightly clustered concepts. Component ablations in Appendix D.1 further confirm the role of multiplicative steering, concept profiles, and the adaptive constraint.

5.2.2 Exp-II: Robustness Under Adversarial Queries

Table 2 evaluates robustness under three query variants: Standard, Injection, and Zou-noise. The key observation is that hit-based suppression and token-budgeted context exposure can diverge: reducing target hits does not always prevent target documents from entering the RAG context.

Hit-based suppression and context exposure can diverge. Across 5 datasets, 6 methods, and 3 attacks, we observe 28 out of 90 configurations where Hit Delta improves but CER Delta worsens. This occurs most often for Random and INLP, indicating that aggressive perturbations and fixed linear projections can reduce top-ranked target hits while still allowing target evidence into the budgeted context. In contrast, C^2U shows only one such case, on Amazon under Injection, where the CER increase is small. This suggests better alignment between ranking-level suppression and context-level safety.

Baseline behavior under attacks. Linear methods often preserve utility but can be sensitive to adversarial query shifts. For example, LEACE reduces Amazon exposure under Standard queries but increases exposure under Zou-noise, despite unchanged utility. Perturbation baselines can reduce hits more aggressively, especially on ArXiv, but they often do so with large utility loss and unstable exposure behavior. On MNLI, DARE-GRL produces large positive CER Delta under all three query variants despite negative Hit Delta, showing that hit suppression does not necessarily imply safer contexts.

Robustness of C^2U . C^2U more consistently controls both target hits and context exposure across attack types. On ArXiv, it maintains strong suppression under Injection and Zou-noise while keeping utility drops in the single digits. On MNLI, it is the only method with non-positive CER Delta across all query variants, avoiding the exposure increases observed for DARE-GRL and Random. TOFU remains challenging because target and non-target evidence are tightly coupled around author identifiers. The TOFU results show that target leakage can often be reduced, but utility degradation can be large; Appendix D.6 further attributes this behavior to non-target spillover rather than residual target leakage.

Overall, Exp-II shows that adversarial robustness cannot be characterized solely by Hit Delta. Token-budgeted exposure is necessary for distinguishing targeted suppression from unsafe or overly broad perturbation. Across datasets and attacks, C^2U more consistently aligns hit suppression with reduced context exposure, while linear projections can be attack-sensitive and perturbation baselines can over-edit the retrieval neighborhood.

5.2.3 Additional Ablations and Operating Regime

Appendix D further characterizes the operating regime of C^2U . Dictionary size and edited atom count control the suppression–utility trade-off (Appendix D.2); profile size and contamination analyses show

Table 2: **Attack-aware Retrieval Unlearning Results (Compact)**. We report performance under Standard, Injection, and Zou-noise attacks. Metrics reported are percentage point (pp) changes: *Hit* Δ ($\downarrow$, lower is better unlearning), *CER* Δ ($\downarrow$, lower is safer), and *Util Drop* ($\downarrow$, lower is better utility preservation).

Dataset	Method	Standard			Injection			Zou-noise		
		Hit Δ	CER Δ	Util $\downarrow$	Hit Δ	CER Δ	Util $\downarrow$	Hit Δ	CER Δ	Util $\downarrow$
Amazon	C ² U	<u>-50.0</u>	<u>-20.6</u>	0.0	<u>-1.0</u>	<u>+0.2</u>	0.0	-23.0	-6.4	0.0
	INLP	<u>-27.0</u>	<u>+6.8</u>	0.0	<u>+3.0</u>	<u>+5.4</u>	0.0	<u>-12.0</u>	<u>+0.4</u>	0.0
	LEACE	-26.0	<u>-5.6</u>	0.0	+7.0	+2.4	0.0	-2.0	+12.6	0.0
	PISCES	-10.0	-2.0	0.0	0.0	0.0	0.0	-2.0	+0.4	0.0
	DARE-GRL	-58.0	-25.4	0.0	-2.0	0.0	0.0	-23.0	-6.4	0.0
	Random	-20.0	+30.8	0.0	+35.0	+57.4	0.0	+15.0	+49.8	0.0
ArXiv	C ² U	-28.6	-3.0	<u>0.7</u>	-90.4	-3.6	9.0	-91.4	-4.1	9.7
	INLP	-5.8	+1.3	<u>0.6</u>	-21.6	+5.7	5.2	-15.0	+8.6	4.0
	LEACE	-6.2	<u>-1.8</u>	0.0	-33.8	<u>-1.5</u>	<u>0.2</u>	-17.2	<u>-1.4</u>	<u>0.2</u>
	PISCES	-0.2	-0.2	<u>0.2</u>	-3.0	+0.9	0.1	-0.2	+1.6	0.1
	DARE-GRL	<u>-79.0</u>	-3.0	<u>69.8</u>	-73.4	-3.6	67.7	-75.8	-4.1	68.8
	Random	-82.4	+7.5	70.3	<u>-78.2</u>	+6.8	69.6	<u>-78.4</u>	+7.6	71.0
IMDB	C ² U	-100.0	-10.6	0.0	-8.0	-6.6	-11.2	-53.0	-7.8	-3.0
	INLP	<u>-88.0</u>	<u>+14.6</u>	43.4	-8.0	-6.6	52.0	-53.0	-7.8	60.2
	LEACE	-100.0	-10.6	<u>8.4</u>	-8.0	+23.6	3.0	-26.0	<u>+0.6</u>	<u>-0.4</u>
	PISCES	0.0	+19.8	0.0	+28.0	+25.8	10.2	+44.0	+34.4	1.0
	DARE-GRL	0.0	+89.4	0.0	+92.0	+93.4	-11.2	+47.0	+92.2	-3.0
	Random	-67.0	+16.0	14.8	<u>+22.0</u>	<u>+19.4</u>	<u>-0.6</u>	<u>-31.0</u>	+17.2	10.4
MNLI	C ² U	<u>-61.0</u>	-18.8	3.8	-5.0	-2.6	3.4	-3.0	-0.2	7.6
	INLP	-9.0	+23.2	1.2	-1.0	<u>-1.6</u>	<u>-2.8</u>	+5.0	+1.8	<u>0.0</u>
	LEACE	-23.0	<u>-6.8</u>	0.4	<u>-3.0</u>	<u>-0.2</u>	1.0	<u>-2.0</u>	<u>0.0</u>	<u>0.0</u>
	PISCES	0.0	+1.2	<u>0.0</u>	+1.0	+0.4	1.0	0.0	+0.4	0.8
	DARE-GRL	-90.0	+80.2	-0.2	-5.0	+97.4	-6.8	-3.0	+99.8	-2.4
	Random	<u>-65.0</u>	+49.8	20.0	+21.0	+69.2	12.6	+18.0	+72.6	17.0
TOFU	C ² U	-85.0	-3.2	70.2	0.0	0.0	7.2	-20.0	-2.0	20.0
	INLP	-85.0	-3.2	74.6	0.0	0.0	8.6	-20.0	-2.0	22.4
	LEACE	<u>-80.0</u>	<u>-3.0</u>	<u>8.2</u>	0.0	0.0	1.0	-20.0	-2.0	7.0
	PISCES	0.0	+20.0	3.8	<u>+25.0</u>	+8.2	<u>1.6</u>	<u>+55.0</u>	+26.4	8.4
	DARE-GRL	-85.0	+96.8	77.0	0.0	+100.0	9.0	-20.0	+98.0	22.6
	Random	-85.0	+1.8	79.6	0.0	<u>+5.6</u>	9.6	-20.0	<u>+0.6</u>	23.2

Note: **Bold** and underline denote best and second-best results (computed per dataset/attack/metric). *CER* (Context Exposure Rate) measures the inclusion of target content in a budgeted retrieval context. Negative Δ values indicate improvement (better unlearning/safety); positive *Util Drop* indicates degradation.

that concept profiles are stable in the tested range (Appendix D.3); and multi-concept experiments show substantial suppression for 2–5 simultaneous targets with increasing utility cost for larger policy sets (Appendix D.7). We also analyze backbone transfer, TOFU entity-centric behavior, cross-domain transfer with optional dictionary extension, and measured deployment overhead in Appendices D.8, D.6, D.9, and D.10.

6 Conclusion

We presented C²U, a retrieval-side framework for concept unlearning in RAG that suppresses target-bearing evidence through inference-time query editing. By learning a sparse atom-space representation, C²U estimates concept profiles and steers target-aligned query components without retriever-backbone retraining or index rebuilding. Across standard and adversarial settings, C²U provides favorable suppression–utility trade-offs, especially for semantically distributed concepts where linear erasure under-suppresses and aggressive perturbation over-edits. Our analyses further show that token-budgeted exposure is necessary for evaluating RAG unlearning, dictionary granularity controls strength versus selectivity, entity-centric concepts require conservative steering, and lightweight dictionary extension provides a practical path for occasional domain adaptation.

References

- [1] Bang An, Shiyue Zhang, and Mark Dredze. Rag llms are not safer: A safety analysis of retrieval-augmented generation for large language models. *arXiv preprint arXiv:2504.18041*, 2025.
- [2] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. 2024.
- [3] M Bahraminasr and A Vafaei-Sadr. Imdb data from two generations (1979 to 2019). part one: Dataset. *CoRR*, 2020.
- [4] Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form. *Advances in Neural Information Processing Systems*, 36:66044–66063, 2023.
- [5] Anh Bui, Trang Vu, Long Vuong, Trung Le, Paul Montague, Tamas Abraham, Junae Kim, and Dinh Phung. Fantastic targets for concept erasure in diffusion models and where to find them. *arXiv preprint arXiv:2501.18950*, 2025.
- [6] Min Chen, Weizhuo Gao, Gaoyang Liu, Kai Peng, and Chen Wang. Boundary unlearning: Rapid forgetting of deep networks via shifting the decision boundary. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7766–7775, 2023.
- [7] Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblani, Dominik Krzemiński, Genta Indra Winata, et al. Mmteb: Massive multilingual text embedding benchmark. *arXiv preprint arXiv:2502.13595*, 2025.
- [8] Chongyu Fan, Jiancheng Liu, Yihua Zhang, Eric Wong, Dennis Wei, and Sijia Liu. Salun: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation. *arXiv preprint arXiv:2310.12508*, 2023.
- [9] Rohit Gandikota, Joanna Materzynska, Jaden Fiotto-Kaufman, and David Bau. Erasing concepts from diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2426–2436, 2023.
- [10] Yaroslav Ganin and Victor Lempitsky. Unsupervised domain adaptation by backpropagation. In *International conference on machine learning*, pages 1180–1189. PMLR, 2015.
- [11] Yoav Gur-Arieh, Clara Haya Suslik, Yihuai Hong, Fazl Barez, and Mor Geva. Precise in-parameter concept erasure in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 18997–19017, 2025.
- [12] Sebastian Hofstätter, Jiecao Chen, Karthik Raman, and Hamed Zamani. Fid-light: Efficient and effective retrieval-augmented text generation. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1437–1447, 2023.
- [13] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, et al. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.
- [14] Xun Liang, Simin Niu, Zhiyu Li, Sensen Zhang, Hanyu Wang, Feiyu Xiong, Zhaoxin Fan, Bo Tang, Jihao Zhao, Jiawei Yang, et al. Saferag: Benchmarking security in retrieval-augmented generation of large language model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4609–4631, 2025.
- [15] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- [16] Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C Lipton, and J Zico Kolter. Tofu: A task of fictitious unlearning for llms. *arXiv preprint arXiv:2401.06121*, 2024.

- 357 [17] Julian McAuley and Jure Leskovec. Hidden factors and hidden topics: understanding rating
358 dimensions with review text. In *Proceedings of the 7th ACM conference on Recommender*
359 *systems*, pages 165–172, 2013.
- 360 [18] Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. Mteb: Massive text
361 embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the*
362 *Association for Computational Linguistics*, pages 2014–2037, 2023.
- 363 [19] Hadas Orgad, Bahjat Kawar, and Yonatan Belinkov. Editing implicit assumptions in text-
364 to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on*
365 *Computer Vision*, pages 7053–7061, 2023.
- 366 [20] Huachuan Qiu, Shuai Zhang, Anqi Li, Hongliang He, and Zhenzhong Lan. Latent jailbreak: A
367 benchmark for evaluating text safety and output robustness of large language models. *arXiv*
368 *preprint arXiv:2307.08487*, 2023.
- 369 [21] Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out:
370 Guarding protected attributes by iterative nullspace projection. *arXiv preprint arXiv:2004.07667*,
371 2020.
- 372 [22] Ayush Kumar Tarun, Vikram Singh Chundawat, Murari Mandal, and Mohan Kankanhalli. Deep
373 regression unlearning. In *International Conference on Machine Learning*, pages 33921–33939.
374 PMLR, 2023.
- 375 [23] Shang Wang, Tianqing Zhu, Dayong Ye, and Wanlei Zhou. When machine unlearning meets
376 retrieval-augmented generation (rag): Keep secret or forget knowledge? *IEEE Transactions on*
377 *Dependable and Secure Computing*, 2025.
- 378 [24] Adina Williams, Nikita Nangia, and Samuel Bowman. A broad-coverage challenge corpus
379 for sentence understanding through inference. In *Proceedings of the 2018 conference of the*
380 *North American chapter of the association for computational linguistics: human language*
381 *technologies, volume 1 (long papers)*, pages 1112–1122, 2018.
- 382 [25] Jing Wu, Trung Le, Munawar Hayat, and Mehrtash Harandi. Erasing undesirable influence in
383 diffusion models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*,
384 pages 28263–28273, 2025.
- 385 [26] Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented
386 generation. 2024.
- 387 [27] Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super
388 mario: Absorbing abilities from homologous models as a free lunch. In *Forty-first International*
389 *Conference on Machine Learning*, 2024.
- 390 [28] Yue Yu, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi,
391 and Bryan Catanzaro. Rankrag: Unifying context ranking with retrieval-augmented generation
392 in llms. *Advances in Neural Information Processing Systems*, 37:121156–121184, 2024.
- 393 [29] Tianzhe Zhao, Jiaoyan Chen, Yanchi Ru, Haiping Zhu, Nan Hu, Jun Liu, and Qika Lin.
394 Exploring knowledge poisoning attacks to retrieval-augmented generation. *Information Fusion*,
395 page 103900, 2025.
- 396 [30] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson.
397 Universal and transferable adversarial attacks on aligned language models. *arXiv preprint*
398 *arXiv:2307.15043*, 2023.

Appendix Overview

This appendix provides additional details to support reproducibility and interpretation of our results. Appendix A specifies the experimental protocol, index construction, evaluation metrics, and the optimization objective used to train ACR. Appendix B describes each dataset, including the target concept, control attribute, query/document construction, and preprocessing. Appendix C documents the implementation details of all baselines under the same default retrieval backbone and evaluation pipeline. Appendix D provides additional analyses of the operating regime of C²U, including component ablations, steering and dictionary granularity, concept-profile robustness, structured atom usage, visual diagnostics, entity-centric behavior, multi-concept suppression, backbone transfer, cross-domain transfer with optional dictionary extension, and deployment cost.

A Experimental Details

A.1 Implementation and Data Protocol

Retrieval Framework. Unless otherwise specified, experiments use the sentence-transformers/all-MiniLM-L6-v2 backbone as the default retriever for controlled comparison. Retrieval similarity is computed as the dot product of L_2 -normalized embeddings. To isolate the impact of representation editing, all baselines and C²U share the same default evaluation pipeline. Additional backbone-transfer experiments with gte-base, bge-base-en-v1.5, and e5-base-v2 are reported in Appendix D.8.

Index Construction. We evaluate performance on per-target sub-indices. For each target concept t , the evaluation index $\mathcal{D}_{\text{eval}}$ is constructed by sampling $N_{\text{tgt}} = 100$ documents containing the target concept and $N_{\text{ctrl}} = 2,000$ control documents. These sample sizes are fixed across all datasets to maintain a consistent imbalance ratio of 1 : 20. For datasets containing multiple concepts, we report average metrics over the selected concepts using deterministic seeds.

Hyperparameters. We report Hit@5 for all retrieval tasks. C²U introduces two dataset-specific inference hyperparameters: the concept truncation size K_{atom} and the steering strength λ . Unless otherwise specified, each concept profile is constructed from 20 associated documents. Profile-size and profile-contamination sensitivity are reported in Appendix D.3, and additional sensitivity to atom sparsity and dictionary size is reported in Appendix D.2. Table 3 details the dataset-specific configurations used in the main experiments.

Dataset	Truncation (K_{atom})	Steering (λ)
Amazon	250	3
ArXiv	30	2
IMDB	100	40
MNLI	50	2
TOFU	100	2

Table 3: Hyperparameter configurations for each dataset. K_{atom} denotes the number of atoms retained in the concept profile, and λ controls the steering magnitude. Unless otherwise specified, concept profiles are constructed from 20 associated documents. Index sizes are fixed to $N_{\text{tgt}} = 100$ and $N_{\text{ctrl}} = 2,000$ for all main experiments.

A.2 Evaluation Metrics

Let $\mathcal{Q}_t$ and $\mathcal{Q}_c$ denote the sets of target queries and control queries, respectively. For a given query q , $\text{TopK}(q)$ represents the set of labels or categories associated with the top- K retrieved documents. We define metric changes relative to the unedited baseline as $\Delta = \text{Metric}_{\text{Method}} - \text{Metric}_{\text{Base}}$.

Erasure and Utility in the Standard Setting. We employ four metrics to assess performance in the absence of adversarial attacks:

434 1. **Target Hit@K Drop** ($\uparrow$) measures the reduction in retrieval success for target queries:

$$\text{Hit@K Drop} = \frac{1}{|\mathcal{Q}_t|} \sum_{q \in \mathcal{Q}_t} (\mathbb{I}[t \in \text{TopK}_{\text{Base}}(q)] - \mathbb{I}[t \in \text{TopK}_{\text{Method}}(q)]). \quad (8)$$

435 2. **Target Cos Drop** ($\uparrow$) quantifies the reduction in cosine similarity between target-query
436 representations and the target concept centroid. Let μ_t be the centroid of all target document
437 embeddings in the base index:

$$\Delta_{\cos} = \frac{1}{|\mathcal{Q}_t|} \sum_{q \in \mathcal{Q}_t} (\cos(\mathbf{v}_{\text{base}}(q), \mu_t) - \cos(\mathbf{v}_{\text{method}}(q), \mu_t)). \quad (9)$$

438 3. **Control/Neighbor Delta** ($\downarrow$) evaluates the impact on non-target retrieval utility. Let y_q be
439 the ground-truth control label for $q \in \mathcal{Q}_c$:

$$\text{Utility Delta} = \frac{1}{|\mathcal{Q}_c|} \sum_{q \in \mathcal{Q}_c} (\mathbb{I}[y_q \in \text{TopK}_{\text{Base}}(q)] - \mathbb{I}[y_q \in \text{TopK}_{\text{Method}}(q)]). \quad (10)$$

440 4. **Target Exposure** Δ ($\downarrow$) measures the change in false-positive retrieval of target documents
441 by control queries:

$$\text{Exposure } \Delta = \frac{1}{|\mathcal{Q}_c|} \sum_{q \in \mathcal{Q}_c} (\mathbb{I}[t \in \text{TopK}_{\text{Method}}(q)] - \mathbb{I}[t \in \text{TopK}_{\text{Base}}(q)]). \quad (11)$$

442 **Robustness under Attacks.** We evaluate robustness under three query variants: *Standard*, *Injection*,
443 and *Zou-noise*. Results are reported as percentage point changes.

- 444 • **TargetHit** Δ (**pp**) ($\downarrow$): the change in retrieval rate of target documents for target queries,
445 defined analogously to Hit@K.
- 446 • **CER** Δ (**pp**) ($\downarrow$): the change in *Context Exposure Rate*. CER measures whether target
447 documents appear in the RAG context constructed from retrieved documents under a token
448 budget $B = 512$. Let $\text{Docs}(q; B)$ denote the set of retrieved documents packed into the
449 context for query q whose cumulative token count does not exceed B , with partial inclusion
450 of the last document if needed. We define:

$$\text{CER}(q) = \mathbb{I}[\text{Docs}(q; B) \cap \mathcal{D}_t \neq \emptyset], \quad (12)$$

451 where $\mathcal{D}_t$ is the set of target documents. We report CER changes relative to the unedited
452 baseline. Unlike Hit@K, which only checks top- K ranking membership, CER accounts for
453 token-budgeted context construction in RAG.

- 454 • **Utility Drop** (**pp**) ($\downarrow$): the reduction in retrieval accuracy for control queries under the
455 corresponding query variant.

456 A.3 Optimization Setup

457 **Training Dynamics.** To preserve the retrieval geometry of the pre-trained dense retriever, we freeze
458 the backbone encoder parameters throughout ACR training. Optimization is restricted to the atom
459 dictionary $\mathcal{A}$ and the lightweight projection head. We use AdamW with a batch size of 32 and an
460 initial learning rate of 10^{-3} . To improve convergence stability, we employ a dynamic scheduler
461 (ReduceLROnPlateau) that decays the learning rate by a factor of 0.5 when the validation metric
462 stagnates, using patience of one epoch. The model is trained on a 90/10 train-validation split for a
463 maximum of 10 epochs with early stopping.

464 **Hyperparameter Configuration.** The composite loss is balanced via empirically selected coeffi-
465 cients. We set the alignment penalty weight $\lambda_{\text{align}} = 1.0$ to anchor ACR-composed representations
466 to the frozen dense baseline, and the entropy regularization weight $\lambda_{\text{ent}} = 0.02$ to encourage peaked
467 atom-weight distributions. The retrieval margin is set to $m_r = 0.3$. For the attention mechanism, we
468 use temperature $\tau = 0.1$ to sharpen the softmax distribution. Unless otherwise specified, the default
469 dictionary size is $N = 1024$.

Training Objective (ACR). We freeze the transformer backbone and optimize only the atom dictionary and a lightweight projection head. For each triplet (q, d^+, d^-) , ACR outputs a composed Euclidean embedding e_x^{acr} and atom weights $w_x \in \Delta^{|\mathcal{A}|-1}$ for $x \in \{q, d^+, d^-\}$. We L_2 -normalize the composed embeddings and compute cosine similarities by dot product:

$$z_x = \frac{e_x^{\text{acr}}}{\|e_x^{\text{acr}}\|_2 + \varepsilon}, \quad \cos(z_a, z_b) = z_a^\top z_b.$$

Let e_x^{base} denote the frozen dense retriever embedding for input x , and let $\bar{e}_x^{\text{base}} = e_x^{\text{base}} / (\|e_x^{\text{base}}\|_2 + \varepsilon)$. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{tri}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}},$$

where:

(i) Cosine triplet retrieval loss.

$$\mathcal{L}_{\text{tri}} = \max\left(0, m_r - \cos(z_q, z_{d^+}) + \cos(z_q, z_{d^-})\right).$$

(ii) Cosine alignment loss. We encourage each ACR-composed embedding to remain close to the corresponding frozen retriever embedding:

$$\mathcal{L}_{\text{align}} = \sum_{x \in \{q, d^+, d^-\}} \left(1 - \cos(z_x, \bar{e}_x^{\text{base}})\right).$$

(iii) Entropy penalty on atom weights. To obtain sharper atom-weight distributions that facilitate localized concept edits, we penalize entropy:

$$\mathcal{L}_{\text{ent}} = \sum_{x \in \{q, d^+, d^-\}} \left(-\sum_{k=1}^{|\mathcal{A}|} w_{x,k} \log(w_{x,k} + \epsilon)\right).$$

Here, $m_r > 0$ is the triplet margin, and $\lambda_{\text{align}}, \lambda_{\text{ent}} > 0$ are scalar weights.

A.4 Compute Reporting

We report the computational resources used for ACR training, retrieval evaluation, baseline comparison, ablations, and deployment-cost measurements. All reported retrieval-side experiments are executed locally. C²U does not train a large language model, fine-tune a generator, or require retriever-index rebuilding during concept unlearning. For C²U, the retriever backbone is frozen; optimization is restricted to the atom dictionary and lightweight projection/composition modules.

Local Compute Environment. All experiments were run on a single GPU from a workstation with 4 NVIDIA RTX 6000 Ada Generation GPUs, each with 47.37 GB of memory. Although the workstation contains 4 GPUs, each reported run used only one GPU and did not use distributed training, multi-node computation, or multi-GPU parallelism. The workstation has 128 CPU cores and 487.80 GB of RAM. The software environment used Linux 6.8, Python 3.12.0, PyTorch 2.5.1, and CUDA 12.4.

Training and Evaluation Setup. Unless otherwise specified, the default retriever backbone is sentence-transformers/all-MiniLM-L6-v2. Additional backbone-transfer experiments use gte-base, bge-base-en-v1.5, and e5-base-v2. During ACR training, the transformer backbone is frozen and only the atom dictionary and lightweight projection/composition head are optimized. The default dictionary contains 1024 atoms. We use AdamW with batch size 32. The training script allows up to 10 epochs with early stopping; in the reported main experiments, runs typically stop after 1–3 epochs.

For each target concept, the evaluation index contains 100 target documents and 2,000 control documents. Retrieval evaluation uses up to 100 target queries and 500 control queries per target concept. Index encoding uses batch size 64 and maximum sequence length 128. Concept profiles use 20 texts by default. Dataset-specific steering hyperparameters are selected on a validation split as reported in Table 3.

Baseline Compute. All baselines are evaluated under the same retrieval pipeline and single-GPU execution setting. INLP fits iterative linear probes and applies nullspace projections. LEACE applies a closed-form erasure transform. PISCES-style feature ablation masks target-aligned representation directions in the frozen retrieval representation. Random adds calibrated Gaussian perturbations to query embeddings. DARE-GRL is run according to the training-based baseline described in Appendix C. All baseline outputs are evaluated using the same fixed-index retrieval protocol.

Local Runtime. Measured deployment latency is reported in Appendix D.10. In the ArXiv setting with 2,100 indexed documents, index construction takes 1047 ms, profile construction takes 60 ms per target concept, and query encoding increases from 1.83 ms for the base retriever to 1.92 ms with C²U, corresponding to a 4.6% query-time overhead. Incremental dictionary-extension runs in Appendix D.9 take 26–65 seconds for representative transfer pairs. Overall, the reported experiments can be reproduced on a single RTX 6000 Ada GPU without distributed training.

Tuning and Ablations. The main hyperparameters, including dictionary size, edited atom count, steering strength, profile size, and contamination level, are studied through fixed validation-selected settings and controlled ablation grids rather than large-scale automated hyperparameter search. Ablation experiments are used to characterize the operating regime of C²U, not to perform model selection on the test sets.

API-Based or Provider-Side Compute. All reported retrieval-side experiments are executed locally on the hardware described above. We do not rely on proprietary API services or provider-side inference for the reported retrieval evaluation, so there is no unobserved provider-side compute in the main experimental results.

B Dataset Details

We evaluate the proposed method across five diverse datasets covering scientific literature, fictional biographies, product reviews, movie reviews, and logical inference. For each dataset, we define the *Target Concept* to be suppressed and the *Control Attribute* to be preserved. Table 4 summarizes the statistics and definitions.

Dataset	Domain	Target Concept (t)	Control Attribute (y)	Query Source
ArXiv	Scientific Papers	Specific Topic	ArXiv Category	Title / Abstract
TOFU	Fictional QA	Sensitive Topic	Author Identity	Natural Question
Amazon	Product Reviews	Sentiment	Product Category	Review Title
IMDB	Movie Reviews	Movie Genre	Sentiment	Synthetic Template
MNLI	Logic Inference	Text Genre	Logic Label	Hypothesis

Table 4: Overview of datasets used in evaluation. The Target Concept represents the semantic information to be suppressed, while the Control Attribute represents the utility signal to be maintained.

Why these datasets. We select these datasets to stress-test concept suppression under heterogeneous retrieval conditions: (i) domain-specific terminology and hierarchical labels (ArXiv), (ii) identity-linked fictional knowledge where topic-level suppression can interact with author identity (TOFU), (iii) real-world sentiment versus topical content confounding (Amazon), (iv) an adversarial query formulation that explicitly mentions the target concept (IMDB), and (v) logic-conditioned matching where the control signal is relational rather than topical (MNLI). Together, they cover both natural and synthetic queries, multiple domains, and distinct target/control factorization patterns.

B.1 Hierarchical Scientific Knowledge (ArXiv)

We construct a hierarchical dataset using the Cornell-University/arxiv corpus to simulate concept suppression in specialized domains. The dataset uses the arXiv taxonomy, where the **Control Attribute** is defined as the primary category (L1, e.g., `cs.CR`, `cs.CV`). The **Target Concept** (L2) is derived via heuristic keyword matching within the abstract, such as mapping terms like “attack” or “malware” to the concept *System Attacks* within the `cs.CR` domain.

For the retrieval task, each document d consists of the concatenated paper title and abstract. Each query q is generated using a stochastic strategy: 50% of queries use the exact paper title to simulate

537 navigational search, while the remaining 50% use the first sentence of the abstract to simulate
 538 semantic search. To ensure dataset balance, we use reservoir sampling to select a maximum of
 539 $N_{\max} = 1,500$ documents per category.

540 B.2 Fictional Knowledge Unlearning (TOFU)

541 The TOFU (Task of Fictional Unlearning) dataset [16] consists of question-answer pairs about fictional
 542 authors, designed for unlearning tasks. We adapt this dataset for retrieval by treating the answer as
 543 the document source. The **Control Attribute** is the Author ID, representing the identity that should
 544 be preserved. The **Target Concept** is the semantic topic of the fact, such as *Family*, *Location*, or
 545 *Work*, which is classified via keyword matching on the question text.

546 The retrieval setup involves documents formatted as biography records, such as “Biography Record
 547 of Author_0...”, and queries taken directly from the dataset questions. This structure tests whether the
 548 retrieval system can suppress topic-level evidence while preserving author-linked retrieval utility.

549 B.3 Classification and Inference Tasks

550 **Product Reviews (Amazon).** Adapted from the `amazon_polarity` dataset, which is derived from
 551 the Amazon Reviews corpus [17] and is also included in text-embedding benchmarks such as
 552 MTEB/MMTEB [18, 7], this task aims to suppress sentiment information while preserving product
 553 categorization. The **Target Concept** is sentiment (Positive/Negative), and the **Control Attribute** is
 554 product category, such as *Books* or *Electronics*, extracted via keyword filtering. Queries are derived
 555 from review titles, while documents contain full review content. This configuration evaluates whether
 556 the model can retrieve documents based on product content independent of emotional tone.

557 **Movie Reviews (IMDB).** In contrast to the Amazon dataset, the IMDB dataset [3] inverts the roles of
 558 sentiment and content. Here, the **Target Concept** is movie genre, such as *Sci-Fi* or *Romance*, and the
 559 **Control Attribute** is sentiment. Since the original dataset lacks titles, we generate synthetic queries
 560 using templates that explicitly mention the target concept, such as “Find a Positive review about
 561 a Sci-Fi movie”. This presents a challenging setting where the query explicitly seeks the concept
 562 intended for suppression.

563 **Logical Inference (MNLI).** Adapted from the `multi_nli` dataset [24], this task evaluates concept
 564 suppression in logical inference contexts. The **Target Concept** is the source text genre, such
 565 as *Government* or *Fiction*, and the **Control Attribute** is the NLI label (*Entailment*, *Neutral*, or
 566 *Contradiction*). The retrieval task is formulated such that the query is the hypothesis and the
 567 document is the premise. The objective is to retrieve premises that hold a specific logical relationship
 568 to the hypothesis, while reducing reliance on the source domain.

569 B.4 Preprocessing Protocol

570 For all datasets, we apply a standardized preprocessing pipeline. Text cleaning includes removal
 571 of LaTeX artifacts for ArXiv, whitespace normalization, and lowercasing for keyword matching
 572 operations. To mitigate class imbalance, we downsample dominant classes to a maximum of
 573 $N = 2,500$ samples, except for ArXiv where $N = 1,500$, and discard classes with fewer than 500
 574 samples. All evaluations are performed on sub-indices constructed from the standard training splits.

575 C Baseline Implementation Details

576 To evaluate C²U, we compare it against a diverse set of unlearning paradigms, ranging from post-hoc
 577 linear subspace projections to training-based representation editing. All baselines are adapted to
 578 the dense retrieval setting using the same default backbone, `all-MiniLM-L6-v2`, to ensure fair
 579 comparison under the main evaluation pipeline.

580 C.1 Linear Concept Erasure Methods

581 These methods operate on the fixed representations produced by the frozen backbone. They rely on
 582 the assumption that target concept information can be reduced through linear subspace operations.

INLP (Iterative Nullspace Projection). Proposed by Ravfogel et al. [21], INLP is an iterative procedure that removes information by projecting representations into the nullspace of concept classifiers. At each iteration, a linear probe is trained to predict the target concept from embeddings. The representations are then projected onto the nullspace of the probe’s weight matrix, reducing linearly recoverable concept information. We perform 30 iterations for all datasets.

LEACE (Least-Squares Concept Erasure). Belrose et al. [4] introduced LEACE as a closed-form solution for concept erasure. Unlike the iterative nature of INLP, LEACE computes a single affine transformation that minimizes dependence between representations and target concept labels under its modeling assumptions. In our retrieval setting, LEACE serves as a strong linear erasure baseline that edits representations without retraining the retriever backbone.

C.2 Model Editing and Fine-Tuning Baselines

These methods involve updating model parameters or applying parameter-space transformations to suppress target concept information while preserving retrieval utility.

PISCES [11]. Originally proposed for LLMs, PISCES uses sparse-autoencoder-style feature decomposition to identify and ablate features correlated with a target concept. We adapt this feature-ablation paradigm to dense retrieval. Specifically, we treat the output projection layer of the retriever as the editable component, identify directions with high cosine similarity to the target concept centroid, and mask them to suppress concept-aligned retrieval behavior while keeping the encoder backbone frozen.

DARE-GRL. We construct a strong training-based baseline combining Gradient Reversal Layers (GRL) [10] with DARE-style model merging [27]. First, we fine-tune a copy of the retriever using a GRL objective: the encoder is trained to reduce concept-classifier predictability while preserving retrieval performance on control data. To mitigate broad parameter drift, we subsequently apply DARE (Drop And REscale), which sparsifies the delta weights by setting 90% of small updates to zero and rescales the remaining updates before merging them back into the base model.

C.3 Naive Baseline

Random Perturbation. To verify that suppression from C²U is due to targeted atom-space steering rather than generic representation degradation, we include a random noise baseline. We inject isotropic Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ into query embeddings. The noise scale σ is calibrated to match the average L_2 norm of the steering vectors produced by C²U, ensuring a fair comparison of perturbation magnitude.

D Additional Experimental Analysis

This section provides additional analyses that characterize the operating regime of C²U. We first isolate the contribution of the main method components, then study steering granularity, concept-profile robustness, structured atom usage, dataset-specific behavior, multi-concept suppression, backbone transfer, cross-domain transfer, optional dictionary extension, and deployment cost.

Unless otherwise specified, the default supplementary setting is ArXiv with dictionary size $N = 1024$, edited atom count $K_{\text{atom}} = 30$, steering strength $\lambda = 2.0$, and $n_{\text{profile}} = 20$ profile documents. Across tables, Hit@K Drop measures target suppression, where larger is better; Control Delta measures non-target degradation, where smaller is better; Cos Drop measures reduction in target cosine similarity, where larger is better; and Exposure Delta measures change in target exposure for non-target queries, where smaller is better.

D.1 Component Ablation

Table 5 dissects the contribution of key components in C²U. We analyze the impact of the multiplicative steering operator, profile guidance, and adaptive constraints. Hit@K Drop measures the reduction of target-document retrieval, where higher values indicate stronger suppression, while

Neighbor Δ quantifies distortion of the local retrieval neighborhood, where lower values indicate better structural preservation. We omit *w/o Renorm* because, in our implementation, the adaptive constraint and re-normalization are coupled: when the adaptive brake is inactive, re-normalization has no effect; when active, both mechanisms jointly regulate the edit magnitude.

Table 5: **Component ablation.** We compare the full C²U model against variants: removing the steering operator (*w/o MulOp*), using random profiles (*w/ RandProf*), and removing adaptive constraints (*w/o Adaptive*). *w/o Renorm* is omitted because it behaves identically to *w/o Adaptive* in our implementation.

Dataset	Variant	Hit@K Drop $\uparrow$	Neighbor Δ $\downarrow$
Amazon	Full Model	0.500	-0.039
	w/o MulOp	0.000	0.000
	w/ RandProf	0.300	-0.034
	w/o Adaptive	0.500	-0.039
ArXiv	Full Model	0.286	0.024
	w/o MulOp	0.000	0.000
	w/ RandProf	0.006	0.004
	w/o Adaptive	0.286	0.024
IMDB	Full Model	1.000	0.032
	w/o MulOp	0.000	-0.050
	w/ RandProf	0.000	0.016
	w/o Adaptive	1.000	0.032
MNLI	Full Model	0.610	0.098
	w/o MulOp	0.010	0.000
	w/ RandProf	-0.020	0.023
	w/o Adaptive	0.610	0.098
TOFU	Full Model	0.350	0.068
	w/o MulOp	0.000	-0.003
	w/ RandProf	0.100	0.057
	w/o Adaptive	0.850	0.612

Necessity of the steering operator. Removing the multiplicative operator (*w/o MulOp*) results in a near-total loss of efficacy across all datasets. Hit@K Drop falls to approximately zero, and Neighbor Δ remains negligible. This confirms that element-wise atom re-weighting is the primary mechanism of our approach; without it, the query representation remains effectively unchanged.

Role of profile guidance. Replacing the target profile with a random vector (*w/ RandProf*) substantially degrades performance. On ArXiv and IMDB, Hit@K Drop collapses to near zero, indicating that suppression in the full model is directionally specific to the concept profile rather than an artifact of random perturbation. On Amazon, partial suppression remains even with random profiles, suggesting that the target set is less geometrically isolated and can be affected by generic direction noise; however, targeted profiles consistently yield stronger suppression.

Impact of adaptive constraints. The adaptive mechanism regulates the suppression–utility trade-off, most visibly on TOFU. Without the adaptive constraint, suppression becomes much stronger but causes large neighborhood distortion. In the full model, the adaptive component restricts this aggressive shift, reducing suppression but preserving the retrieval neighborhood. On Amazon, ArXiv, and MNLI, the adaptive constraint is less active, indicating that the default steering does not exceed the safety threshold in those settings.

In summary, this ablation confirms that C²U relies on the multiplicative operator for editing capability, the concept profile for directional precision, and the adaptive mechanism as a safety valve for entity-centric or tightly coupled regimes.

D.2 Steering and Dictionary Granularity

We analyze how the suppression–utility trade-off changes with steering strength, the number of edited atoms, and dictionary size. Figure 3 visualizes the performance landscape across five datasets.

Table 6: **Sensitivity to edited atom count on ArXiv.** Increasing the number of edited atoms broadens the edit. Suppression strengthens and eventually saturates, while non-target degradation continues to increase. The default setting uses 30 edited atoms.

K	Ratio	Hit@K Drop $\uparrow$	Control Delta $\downarrow$	Cos Drop $\uparrow$	Exposure Delta $\downarrow$
8	0.0078	0.018	0.0030	0.0044	-0.0576
15	0.0146	0.046	0.0068	0.0111	-0.0776
30	0.0293	0.286	0.0241	0.0296	-0.0980
60	0.0586	0.744	0.0820	0.0706	-0.1076
90	0.0879	0.916	0.1567	0.1062	-0.1088
120	0.1172	0.954	0.2484	0.1317	-0.1088

We define **Safety** as target suppression, measured by Hit@K Drop, and **Utility** as preservation of neighborhood structure, measured by the negative Neighbor Drop. Darker colors indicate stronger performance for the corresponding metric.

Impact of steering strength and edited atom set. Performance is governed jointly by the steering strength λ and the number of edited atoms K_{atom} . Increasing λ generally strengthens suppression, but overly strong steering can reduce non-target utility. Increasing K_{atom} broadens the intervention: small edited sets produce conservative edits that may under-suppress, while larger edited sets can achieve stronger suppression at the cost of higher non-target degradation. This motivates selecting the mildest setting that satisfies the desired suppression target under a validation utility budget.

Dataset-specific regimes. The response patterns across datasets fall into three representative regimes. Amazon and MNLI are responsive at relatively low steering strengths. ArXiv and TOFU exhibit a smoother trade-off where stronger suppression gradually increases utility cost. IMDB shows a threshold effect, requiring a larger steering strength before suppression becomes effective. These patterns motivate dataset-specific hyperparameter selection in Table 3.

Configuration selection. We select the final hyperparameters by maximizing suppression subject to a utility constraint on a validation split. In practice, we first choose a moderate dictionary size, then tune λ and K_{atom} jointly to meet the desired suppression level while keeping non-target degradation within the allowed budget.

Table 7: **Sensitivity to dictionary size on ArXiv.** Smaller dictionaries produce broader atoms and stronger suppression, while larger dictionaries preserve utility more conservatively. The default dictionary size is 1024.

Dictionary Size	K	Ratio	Hit@K Drop $\uparrow$	Control Delta $\downarrow$	Cos Drop $\uparrow$
256	8	0.03	0.546	0.0646	0.0585
512	15	0.03	0.336	0.0264	0.0483
1024	30	0.03	0.286	0.0241	0.0296
2048	60	0.03	0.120	0.0115	0.0090

Tables 6 and 7 show that atom granularity directly controls the suppression–utility trade-off. Increasing the edited atom count strengthens suppression but also increases collateral degradation. Varying dictionary size changes the granularity of the learned atom space: smaller dictionaries yield coarser but more aggressive edits, while larger dictionaries yield more conservative edits. We therefore treat the default configuration as a validation-selected operating point rather than a universally optimal setting. Figure 4 visualizes the same trend: increasing the edited atom count improves suppression but eventually enters a saturation regime where additional suppression gains are small relative to the increase in non-target degradation.

D.3 Concept-Profile Robustness

This section evaluates whether concept profiles are sensitive to the number and quality of associated documents used for profile construction. Unless otherwise specified, the main experiments use 20 profile documents per target concept.

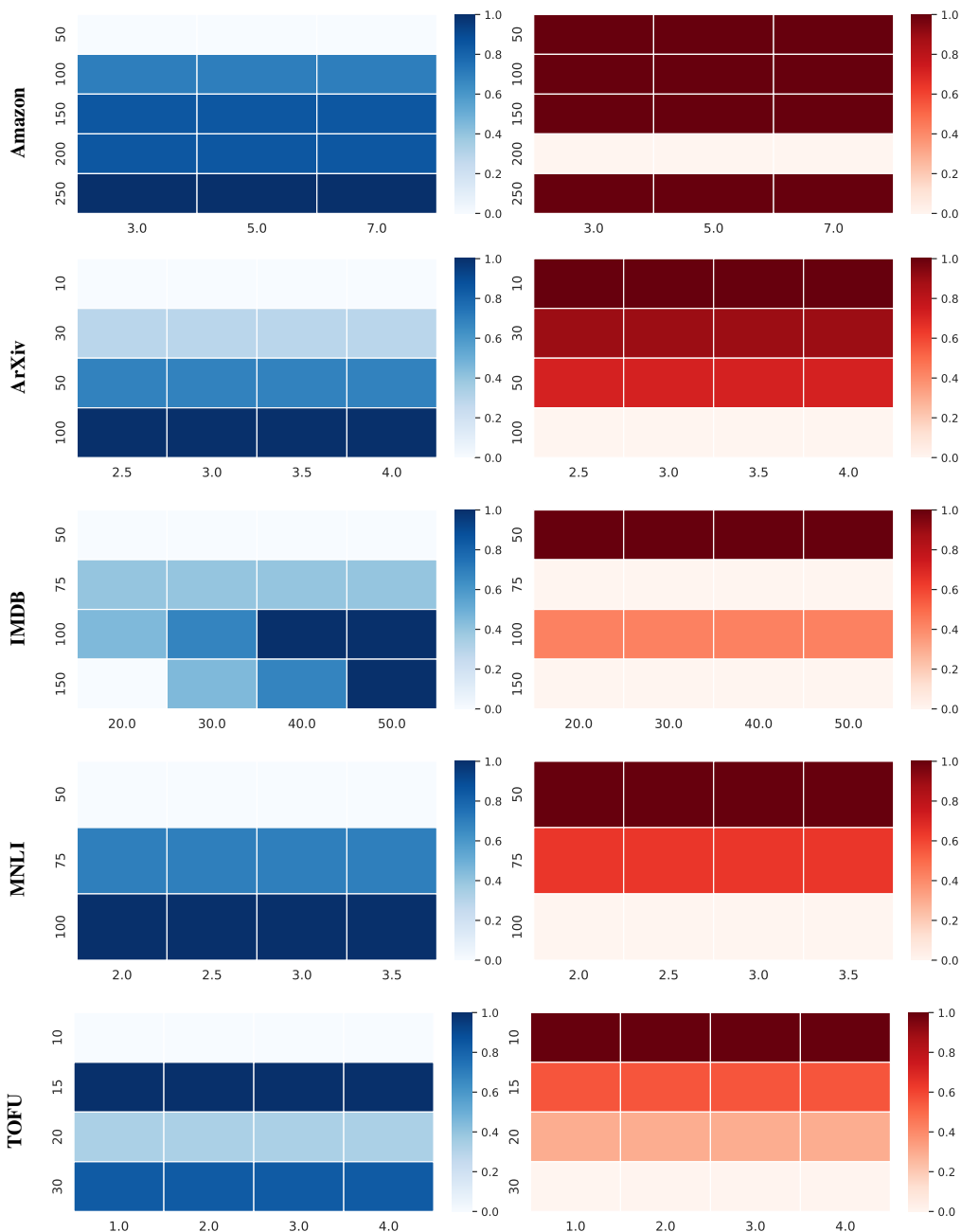


Figure 3: **Safety-utility trade-off across datasets.** Each row displays sensitivity to steering strength λ and edited atom count K_{atom} . **Left:** Safety, where darker blue indicates higher target suppression. **Right:** Utility, where lighter red indicates lower non-target degradation. The results show that moderate steering and validation-selected atom truncation provide balanced operating points.

Profile size. Table 8 shows that increasing the profile size improves suppression with diminishing returns. Increasing the profile size from 5 to 50 raises Hit@K Drop from 0.196 to 0.306, while Control Delta remains stable between 0.0203 and 0.0241. This suggests that C²U does not rely on a fragile profile sample.

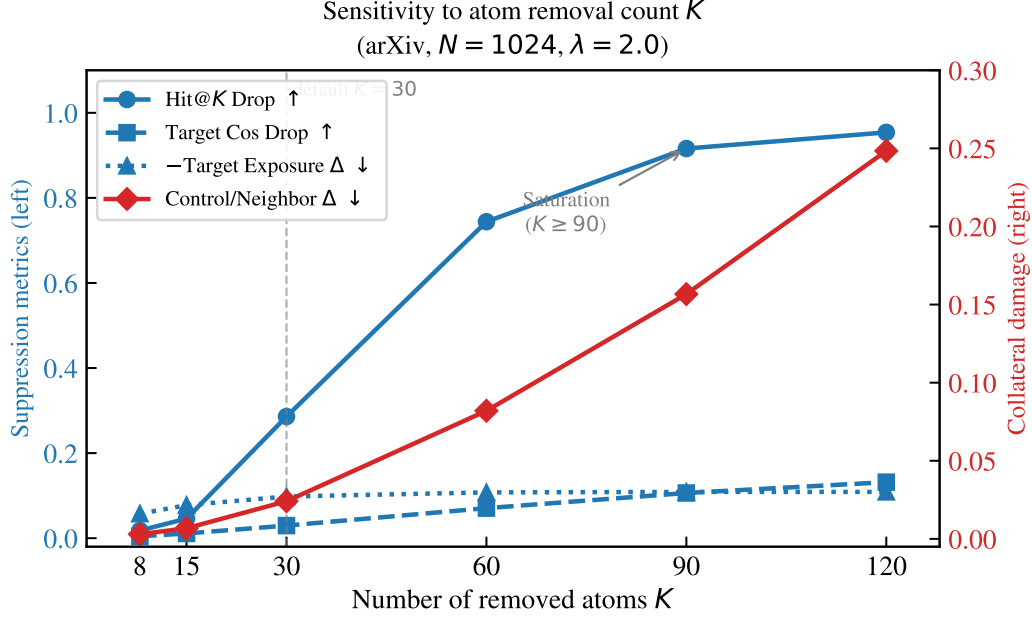


Figure 4: **Sensitivity to edited atom count on ArXiv.** Suppression strengthens as the edited atom count increases and saturates at larger values, while collateral degradation continues to rise. The default setting uses 30 edited atoms, which provides a conservative operating point before the high-degradation regime.

Table 8: **Sensitivity to profile size on ArXiv.** Larger profile sets improve suppression with diminishing returns, while non-target degradation remains stable.

Profile Size	Hit@K Drop $\uparrow$	Control Delta $\downarrow$	Cos Drop $\uparrow$	Exposure Delta $\downarrow$
5	0.196	0.0217	0.0297	-0.0964
10	0.256	0.0224	0.0300	-0.0956
20	0.286	0.0241	0.0296	-0.0980
50	0.306	0.0203	0.0294	-0.1008

689 **Profile contamination.** Table 9 studies profile quality by injecting non-target documents into the
690 profile set. Moderate contamination does not collapse the profile estimate. Instead, contamination
691 appears to broaden the profile direction: Hit@K Drop increases from 0.114 to 0.270, while Control
692 Delta increases mildly from 0.0159 to 0.0226. As visualized in Figure 5, Cos Drop remains nearly
693 flat around 0.029 across contamination levels, suggesting that the main effect is reduced selectivity
694 rather than loss of the target direction. This behavior indicates that noisy profiles are more likely to
695 cause coarser edits than outright failure.

Table 9: **Sensitivity to profile contamination on ArXiv.** Contamination tends to broaden the estimated profile direction: suppression increases, while selectivity decreases mildly.

Noise Ratio	Hit@K Drop $\uparrow$	Control Delta $\downarrow$	Cos Drop $\uparrow$	Exposure Delta $\downarrow$
0.0	0.114	0.0159	0.0292	-0.0784
0.1	0.126	0.0158	0.0280	-0.0864
0.2	0.182	0.0169	0.0286	-0.0908
0.5	0.270	0.0226	0.0294	-0.0956

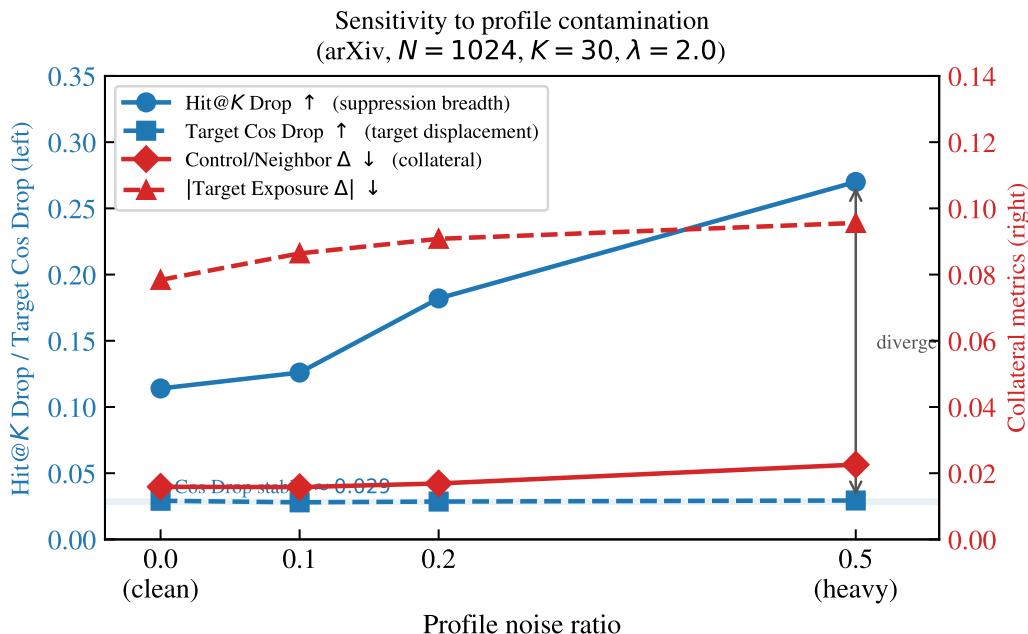


Figure 5: **Profile contamination sensitivity on ArXiv.** Hit@K Drop increases as contamination rises, while Cos Drop remains nearly flat around 0.029. This pattern suggests that contamination broadens the estimated profile direction rather than collapsing it, leading to stronger but less selective suppression.

Overall, small profile sets are sufficient for coarse concept-level suppression, while fine-grained or polysemous concepts may require more carefully curated profiles.

D.4 Evidence for Structured Atom Usage

We analyze the learned atom dictionary to understand whether C²U uses atoms in a structured way. Our goal is not to prove that every atom is a fully disentangled, human-labeled semantic primitive. Instead, we evaluate whether the atom space supports sparse profile estimation, localized steering, and recurring thematic structure.

Sparsity and localized control. The left column of Figure 6 illustrates the histogram of active atoms per document, defined as the number of atoms with attention weights exceeding threshold $\tau = 0.002$. On ArXiv, each document activates 82.8 out of 1024 atoms on average, corresponding to 91.9% sparsity. Across datasets, documents activate only a subset of the dictionary. This sparsity provides operational evidence that the learned atom space supports localized control: compact concept profiles can be estimated and steered without editing the full dense representation. We do not interpret this as proof of fully disentangled semantic primitives.

Atom usage and thematic coherence. Table 10 summarizes quantitative indicators of structured atom usage. High-usage atoms are associated with recurring topical themes, and representative examples are shown in Table 11. These results support an operational view of compositionality: atoms provide reusable coordinates for profile estimation and steering, even if they are not manually labeled semantic primitives.

Table 10: **Atom dictionary analysis on ArXiv.** The learned atom space exhibits sparse usage and partial thematic structure. We interpret these statistics as operational evidence of structured atom usage, not as proof of fully disentangled semantic primitives.

Metric	Value	Interpretation
Active atoms per document	82.8 / 1024	Sparse usage
Sparsity ratio	91.9%	Most atoms inactive per document
Norm-usage correlation	0.280	Modest hierarchy
Top-10 purity	37.0%	Partial consistency
Intra-group cosine	0.1786	Weak cohesion

Table 11: **Representative high-usage atoms on ArXiv.** Labels are post-hoc summaries of high-activation documents and are used only for qualitative interpretation.

Atom ID	Post-hoc Label	Purity	Representative Keywords
492	Genomics	0.60	assembly, datasets, single
485	Biomolecules	0.50	protein, network, security
901	GR and Cosmology	0.50	structures, field, nucleat
564	Biomolecules	0.40	domain, learning, neto2
475	Crypto and Security	0.30	vulnerabilities, software

Cross-dataset patterns. The right column of Figure 6 visualizes the relationship between atom norm and global usage frequency. The positive trend suggests that frequently used atoms tend to occupy more prominent directions in the learned space. This pattern appears across multiple datasets, suggesting that sparse atom usage is recurring. However, this does not imply domain-agnostic generalization; cross-domain transfer and optional dictionary extension are evaluated separately in Appendix D.9.

D.5 Visual Diagnostics for Concept Locality and Steering

To clarify why the same top- K_{atom} steering procedure exhibits different apparent efficacy across tasks, we provide two complementary visual diagnostics. Figure 7 characterizes concept-profile sharpness, while Figure 8 visualizes projection-score distributions before and after steering.

Concept locality via profile sharpness. In Figure 7, we sort atom weights in each concept profile in descending order and plot the normalized weight as a function of atom rank. Sharper drop-offs indicate that the concept is supported by a compact atom subset and is more naturally amenable to top- K_{atom} steering. Long-tailed profiles indicate more distributed support, where suppressing only the top-ranked atoms may yield partial attenuation rather than clean removal.

Separability and intervention effects via projection distributions. Figure 8 visualizes the distribution of projection scores onto the concept direction for *Target (Base)*, *Control*, and *Target (Steered)*. This view shows whether the concept direction separates target from control at baseline and whether steering shifts the target distribution toward the control distribution. For distributed or overlapping concepts, steering may reduce projection magnitude without producing a clean separation pattern.

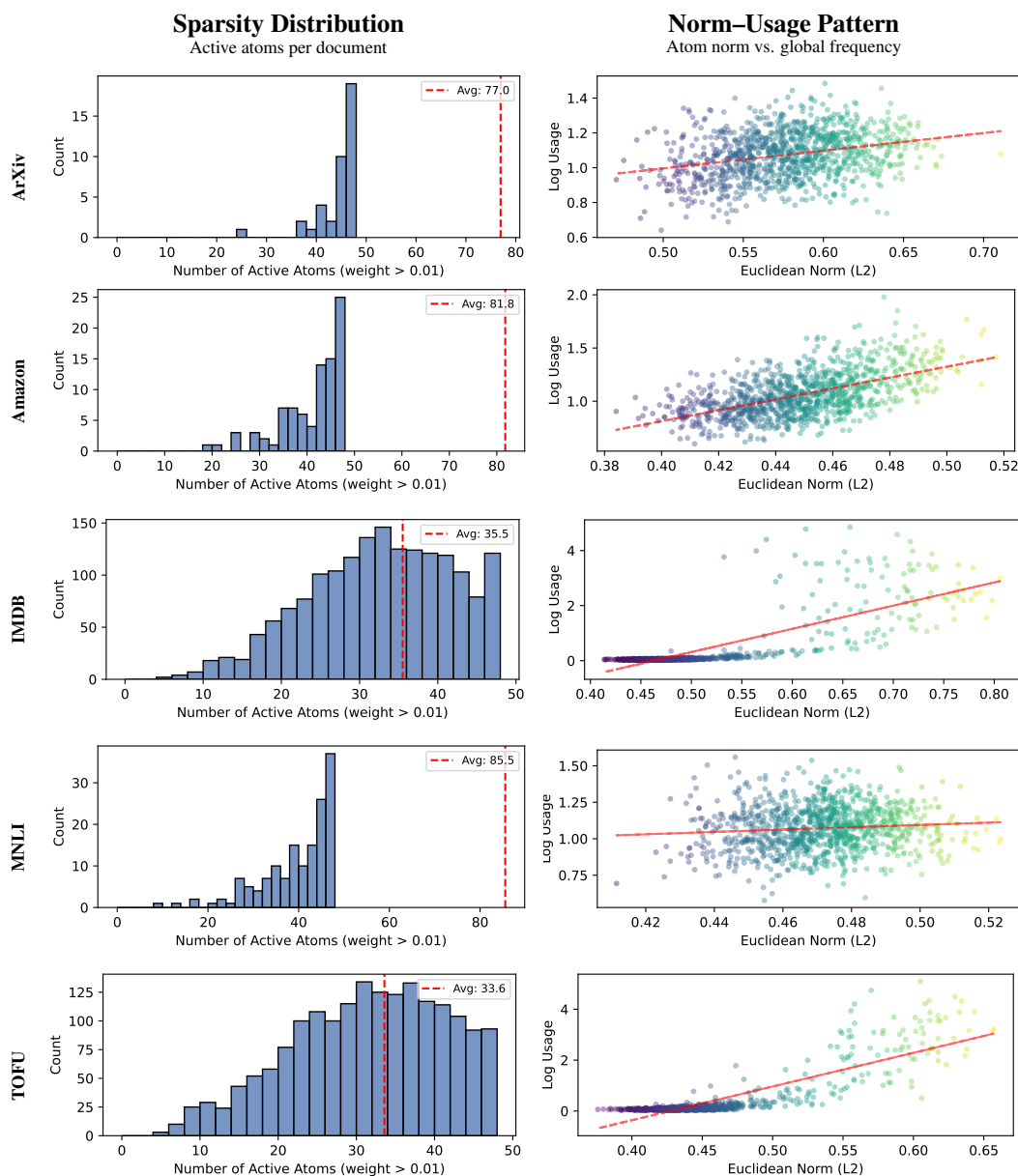


Figure 6: **Atomic usage analysis across datasets.** Each row corresponds to a dataset. **Left:** Histogram of active atoms per document. **Right:** Scatter plot of atom norm against log global usage. These plots show sparse and recurring atom usage patterns, supporting localized steering but not implying fully disentangled semantic primitives.

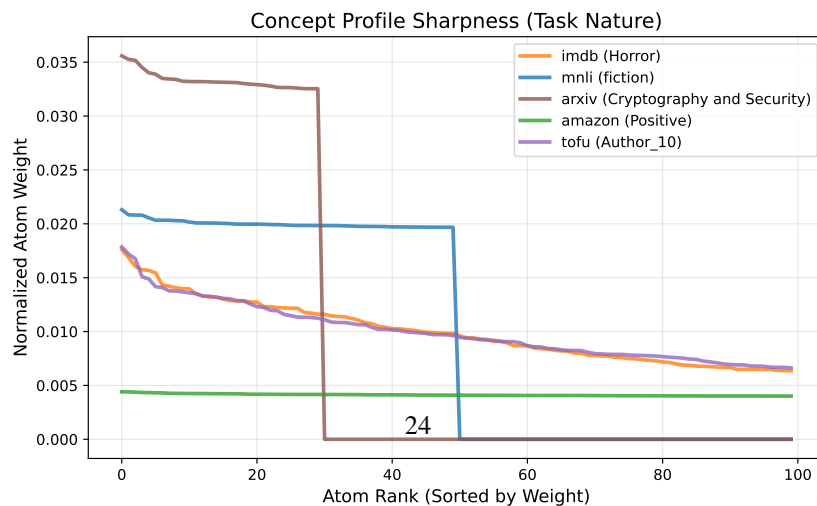


Figure 8: Distribution of normalized atom weights. Sharpness is measured by the slope of the curve at rank 24.

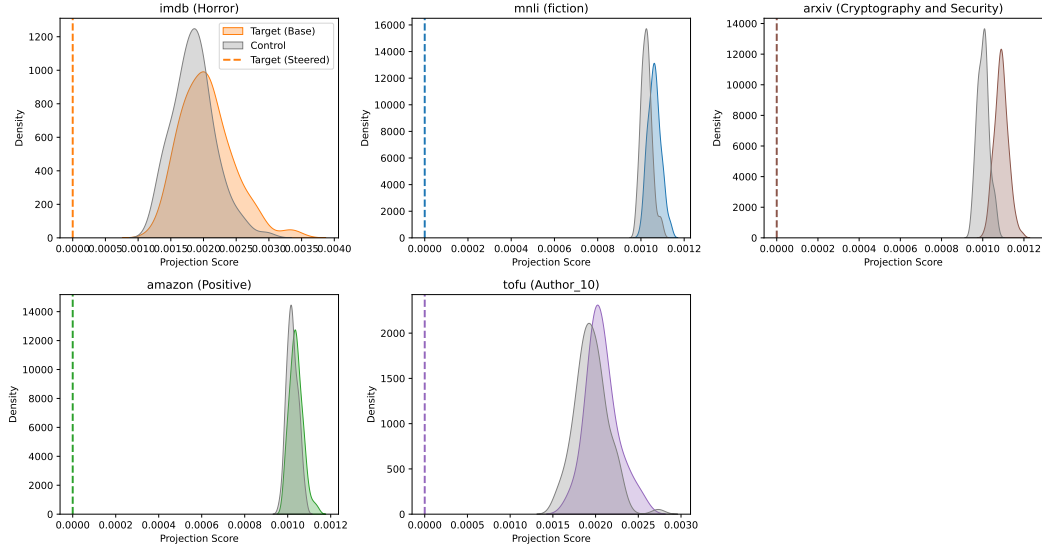


Figure 7: **Concept-profile sharpness across tasks.** Normalized atom weights are sorted in descending order. Sharper drop-offs indicate more localized concepts, while flatter or long-tailed profiles indicate distributed concept support.

Summary. The sharpness plot primarily reflects removability, i.e., how concentrated the concept is over atoms. The projection plot reflects task-level interpretability, i.e., how separable the direction is between target and control and how steering shifts the target distribution. The most visually consistent cases arise when the concept is both localized and baseline-separable.

D.6 Entity-Centric Behavior on TOFU

Table 12 reports a non-default per-query diagnostic setting on TOFU, designed to distinguish residual target leakage from non-target spillover. This diagnostic should not be directly compared with the conservative default setting used in Table 1; instead, it asks what failure mode appears when stronger target suppression is achieved in an entity-centric regime. TOFU is challenging because target and non-target facts are tightly coupled around author identifiers. Under this diagnostic setting, target suppression succeeds and suppression failures are absent, while the main cost is spillover to semantically adjacent non-target queries.

Table 12: **Entity-centric diagnostic on TOFU.** This non-default per-query diagnostic separates residual target leakage from non-target spillover. It shows that, when stronger target suppression is achieved, the dominant failure mode is semantic spillover to adjacent non-target content rather than target leakage.

Metric	Value
Hit@K Drop $\uparrow$	0.850
Cos Drop $\uparrow$	0.0434
Full Suppression $\uparrow$	100.0%
Partial Suppression	0.0%
Suppression Fail $\downarrow$	0.0%
Exposure Delta $\downarrow$	-0.038
Clean Non-target Queries $\uparrow$	27.8%
Degraded Non-target Queries $\downarrow$	72.2%
Collateral Exposure $\downarrow$	0.0%

This analysis indicates that C²U can suppress target evidence in the TOFU diagnostic setting, but doing so can substantially degrade semantically adjacent non-target queries. This explains why the main comparison uses a more conservative default configuration for TOFU: entity-level or author-centric settings require careful steering because target facts and adjacent non-target facts share a tightly coupled retrieval neighborhood.

D.7 Concept-Pool and Multi-Concept Suppression

We evaluate whether C^2U remains stable when the target concept pool expands and when multiple concepts are suppressed simultaneously.

Expanded concept pool. Table 13 reports concept-pool scaling on ArXiv. Expanding the concept pool does not cause monotonic collapse: suppression remains in a stable range while non-target degradation remains small.

Table 13: **Concept-pool evaluation on ArXiv.** Expanding the candidate concept pool does not cause monotonic collapse. Suppression remains in a stable range, and non-target degradation remains small.

Pool	Hit@K Drop $\uparrow$	Control Delta $\downarrow$	Cos Drop $\uparrow$	Exposure Delta $\downarrow$
pool.5	0.2060	0.0114	0.0251	-0.0520
pool.10	0.2580	0.0165	0.0243	-0.0792
pool.15	0.2260	0.0145	0.0266	-0.0872
pool.all	0.2420	0.0159	0.0294	-0.0848

Simultaneous multi-concept suppression. Table ?? evaluates simultaneous suppression of multiple target concepts. Suppression remains substantial across the tested groups. Utility cost remains moderate for smaller groups and increases for the five-concept group, suggesting that larger policy sets should be paired with utility-budgeted calibration.

CV denotes Computer Vision, ML denotes Machine Learning, CL denotes Computation and Language, Prob. denotes Probability, and Gen.Rel. denotes General Relativity. Interference is defined as the average single-concept Hit@K Drop minus the multi-concept Hit@K Drop. Lower values are better; positive values indicate destructive interference, while negative values indicate synergistic suppression relative to the average single-concept baseline.

These results suggest that simultaneous suppression is feasible in the tested regime, but utility-aware calibration becomes more important as the policy set grows.

D.8 Backbone Transfer

We evaluate whether atom-space steering is tied to the default all-MiniLM-L6-v2 retriever. Table ?? reports results on additional encoder backbones. The results indicate that C^2U is not specific to all-MiniLM-L6-v2, but different embedding geometries require different steering calibration. Some backbones permit strong target suppression but with higher utility cost, indicating that backbone transfer is feasible but calibration-dependent.

The selected steering strength is chosen separately for each backbone. The results show that atom-space steering can produce strong suppression beyond all-MiniLM-L6-v2, but the utility cost varies substantially across embedding geometries. The high Control Delta for gte-base and bge-base-en-v1.5 indicates that stronger backbones still require careful calibration. e5-base-v2 shows weaker suppression and positive exposure changes in this setting, possibly due to its query/passage prefix convention. We therefore treat backbone transfer as feasible but calibration-dependent.

D.9 Cross-Domain Transfer and Dictionary Extension

We study whether a trained atom dictionary transfers across domains and whether transfer degradation can be reduced through lightweight dictionary extension.

Zero-shot cross-domain transfer. Figure 9 provides a visual summary of Table ?? . The results show conditional rather than universal transfer. Transfer is strong for some related pairs, such as Amazon to IMDb and MNL to ArXiv. It weakens for larger domain gaps and becomes adverse for several BiosBias transfer pairs. We therefore view the dictionary as reusable within a deployment domain and partially transferable across related domains, rather than domain-agnostic.

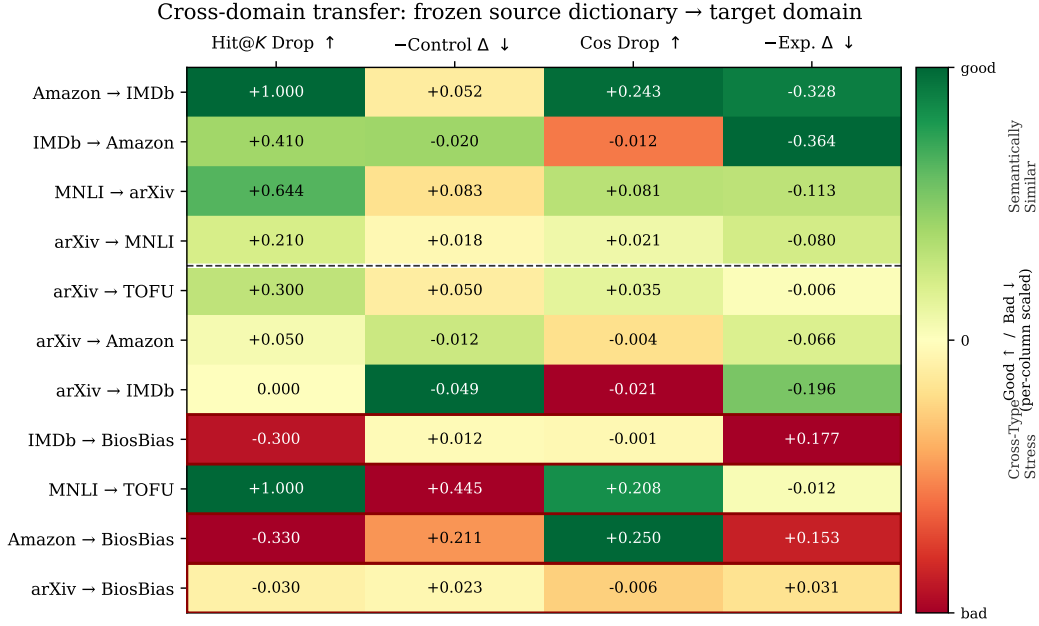


Figure 9: **Zero-shot cross-domain transfer heatmap.** Colors are scaled per metric column, with green indicating more favorable values and red indicating less favorable values. Dark red borders mark adverse-transfer cases with negative Hit@K Drop. These cases are concentrated in BiosBias target transfers, highlighting that the fixed dictionary is not domain-agnostic under large domain shifts.

789 The results show conditional rather than universal transfer. Transfer is strong for some related pairs,
 790 such as Amazon to IMDB and MNLI to ArXiv. It weakens for larger domain gaps and becomes
 791 adverse for several BiosBias transfer pairs. We therefore view the dictionary as reusable within a
 792 deployment domain and partially transferable across related domains, rather than domain-agnostic.

793 **Incremental dictionary-extension ablation.** To test whether domain-shift degradation can be
 794 reduced without full retraining, we conduct an incremental dictionary-extension ablation. Starting
 795 from a trained ACR module, we freeze the original ACR backbone and the existing dictionary, append
 796 64 new trainable atoms, and fine-tune only the query projection layer together with the new atom
 797 parameters. This appends 6.2% additional atoms to the 1024-atom dictionary, updates 0.7% of total
 798 parameters, corresponding to 172K out of 23.3M parameters, and requires no index rebuilding.

Table 17: **Incremental dictionary-extension ablation under domain shift.** We append 64 trainable atoms and update only 0.7% of total parameters. The experiment tests whether transfer degradation can be reduced without full retraining or index rebuilding.

Transfer Pair	Domain Gap	Zero-shot $\uparrow$	+ Incremental $\uparrow$	Gain	Train Time
IMDb $\rightarrow$ Amazon	Low	0.41	0.42	+0.01	44s
ArXiv $\rightarrow$ MNLI	Mid	0.21	0.72	+0.51	65s
IMDb $\rightarrow$ BiosBias	High	-0.30	-0.15	+0.15	26s

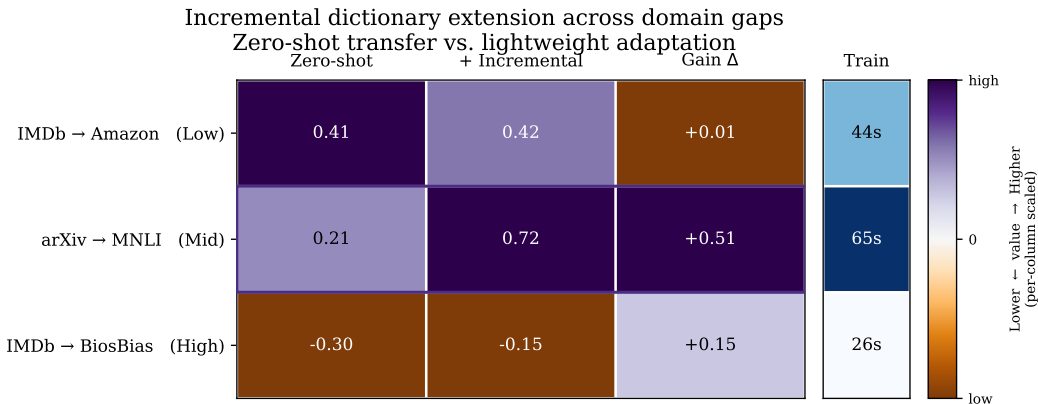


Figure 10: **Incremental dictionary extension across representative transfer pairs.** The figure visualizes zero-shot and incrementally extended performance for low-, mid-, and high-gap transfer pairs. Lightweight extension preserves performance in the low-gap case, substantially improves the mid-gap case, and partially recovers the adverse high-gap case.

Table 17 and Figure 10 show that lightweight extension preserves performance in a low-gap transfer pair, substantially improves a moderate-gap pair, and partially recovers an adverse high-gap pair. We interpret this as preliminary evidence that dictionary adaptation can be performed at low cost when deployment domains shift. We do not claim that this fully solves dynamic open-domain adaptation; rather, it provides an optional mechanism that reduces the cost of adapting the atom space compared with full retraining.

D.10 Efficiency and Deployment Cost

C²U adds a one-time profile construction cost and a small query-time steering cost. Once the dictionary and profiles are available, policy updates are applied through query-side steering without retriever retraining or index rebuilding.

Table 18: **Measured wall-clock cost on ArXiv.** Encode latencies are averaged over 95 queries after discarding 5 warm-up queries. Each pass is timed independently with GPU synchronization to avoid cache interference.

Stage	Mean	Note
Index Build	1047 ms	One-time offline; indexed corpus size is 2100 documents
Profile Construct	60 ms	One-time offline per target concept; profile size is 20 texts
Query Encode Base	1.83 ms	Per query
Query Encode C ² U	1.92 ms	Per query; overhead is +4.6%

In the measured ArXiv setting, C²U increases query encoding time from 1.83 ms to 1.92 ms, adding 0.09 ms per query. Profile construction takes 60 ms per target concept as a one-time offline step. These results support the intended deployment use case: after one-time ACR training and profile construction, concept-level policy updates can be applied through lightweight query-side steering without retriever retraining or index rebuilding.

D.11 Compute Resources

All experiments were run on a single GPU from a workstation with 4 NVIDIA RTX 6000 Ada Generation GPUs, each with 47.37 GB of memory. Although the machine contains 4 GPUs, each reported run used only one GPU and did not use distributed training or multi-GPU parallelism. The workstation has 128 CPU cores and 487.80 GB of RAM. The software environment used Linux 6.8, Python 3.12.0, PyTorch 2.5.1 with CUDA 12.4.

The default retriever backbone is sentence-transformers/all-MiniLM-L6-v2. During ACR training, the transformer backbone is frozen and only the atom dictionary and lightweight projection/composition head are optimized. The default dictionary contains 1024 atoms. The training script

823 supports early stopping; in the reported main experiments, we use dataset-specific training budgets of
824 1–3 epochs with batch size 32.

825 For each target concept, the evaluation index contains 100 target documents and 2,000 control docu-
826 ments. Retrieval evaluation uses up to 100 target queries and 500 control queries per target concept.
827 Index encoding uses batch size 64 and maximum sequence length 128. Baseline methods are evalu-
828 ated under the same pipeline. INLP and PISCES-Lite fit linear SVM probes on index embeddings,
829 LEACE applies a closed-form erasure transform, Random adds calibrated Gaussian perturbations,
830 and DARE-GRL trains a lightweight two-layer mapper/adversary with hidden dimension 512 for 10
831 epochs and batch size 256.

832 Measured deployment latency is reported in Appendix D.10. In the ArXiv setting with 2,100 indexed
833 documents, index construction takes 1047 ms, profile construction takes 60 ms per target concept,
834 and query encoding increases from 1.83 ms for the base retriever to 1.92 ms with C²U, corresponding
835 to a 4.6% query-time overhead. Incremental dictionary-extension runs in Appendix D.9 take 26–65
836 seconds for representative transfer pairs. Overall, the reported experiments can be reproduced on a
837 single RTX 6000 Ada GPU without distributed training.

838 **E Limitations**

839 C²U is designed for embedding-based RAG systems with a fixed retriever index and query-side
840 intervention. Its effectiveness depends on the geometry of the underlying retriever and on the quality
841 of the target concept profile, and different backbones require calibration of the steering strength
842 and edited atom set. Entity-centric regimes such as TOFU are more challenging because target and
843 non-target evidence can be tightly coupled around identifiers, requiring conservative steering to avoid
844 semantic spillover. Our experiments cover five datasets and several adversarial query variants, but do
845 not exhaustively cover all open-domain RAG deployments or multilingual settings.

846 **F Broader Impact**

847 This work aims to reduce unwanted retrieval exposure in RAG systems by suppressing target-bearing
848 evidence before context construction. Potential positive impacts include improved privacy, safety,
849 and policy compliance for retrieval-augmented applications. Potential risks include misuse for
850 hiding evidence, selectively suppressing legitimate information, or enforcing biased content policies.
851 We therefore view C²U as a control mechanism that should be deployed with transparent policy
852 definitions, auditing, and monitoring of non-target utility and exposure.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the main contribution as inference-time, query-side concept unlearning for RAG without retriever retraining or index rebuilding. The experimental results and appendices support the claimed suppression–utility trade-offs, adversarial robustness analysis, and operating-regime characterization.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The paper discusses limitations in Section E and in the operating-regime analyses in Appendix D, including backbone-dependent calibration, entity-centric spillover, profile sensitivity, and domain-shift behavior.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not present formal theoretical results, theorems, or proofs. The mathematical content defines the representation, training objective, and inference-time steering procedure.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper provides the method definition, training objective, dataset construction protocol, evaluation metrics, hyperparameters, baselines, and implementation details in Sections 4–5.1 and Appendices A–C.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We provide anonymized code at <https://anonymous.4open.science/r/C2U-5C88/>, and instructions for reproducing the main experiments, including data pre-processing, ACR training, concept-profile construction, baseline execution, and evaluation scripts. The paper also specifies dataset sources, hyperparameters, and evaluation protocols in Appendix A.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The experimental setup specifies datasets, target/control definitions, metrics, index construction, hyperparameters, optimizer, training schedule, backbone, baselines, and attack variants in Section 5.1 and Appendices A–B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The main tables report deterministic aggregate metrics over fixed evaluation splits and concept sets, but do not include error bars or statistical significance tests. We focus on controlled comparisons under shared retrieval pipelines and provide additional ablations and operating-regime analyses to support the main trends.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix D.11 reports the hardware and software environment, including single-GPU execution on one NVIDIA RTX 6000 Ada GPU, CPU cores, RAM, PyTorch/CUDA versions, batch sizes, sequence length, index size, query counts, and baseline-specific training costs. Appendix D.10 reports measured wall-clock costs for index building, profile construction, query encoding, and incremental dictionary extension.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research uses public datasets and pretrained embedding models, and focuses on reducing unwanted retrieval exposure in RAG systems. We do not conduct human-subject experiments or release high-risk generative models.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section F discusses positive impacts such as privacy, safety, and policy compliance, as well as potential negative uses such as selective suppression of legitimate information or biased policy enforcement.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release a high-risk pretrained generative model, scraped dataset, or deployment system. The method is evaluated as a retrieval-side control mechanism using public datasets and frozen embedding models.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.

- 1114 • We recognize that providing effective safeguards is challenging, and many papers do
1115 not require this, but we encourage authors to take this into account and make a best
1116 faith effort.

1117 12. Licenses for existing assets

1118 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1119 the paper, properly credited and are the license and terms of use explicitly mentioned and
1120 properly respected?

1121 Answer: [No]

1122 Justification: The paper cites the existing datasets and models used in the experiments, but
1123 the current version does not explicitly list the license or terms of use for every asset. We
1124 will verify and document these licenses in the final version.

1125 Guidelines:

- 1126 • The answer [N/A] means that the paper does not use existing assets.
- 1127 • The authors should cite the original paper that produced the code package or dataset.
- 1128 • The authors should state which version of the asset is used and, if possible, include a
1129 URL.
- 1130 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1131 • For scraped data from a particular source (e.g., website), the copyright and terms of
1132 service of that source should be provided.
- 1133 • If assets are released, the license, copyright information, and terms of use in the
1134 package should be provided. For popular datasets, `paperswithcode.com/datasets`
1135 has curated licenses for some datasets. Their licensing guide can help determine the
1136 license of a dataset.
- 1137 • For existing datasets that are re-packaged, both the original license and the license of
1138 the derived asset (if it has changed) should be provided.
- 1139 • If this information is not available online, the authors are encouraged to reach out to
1140 the asset's creators.

1141 13. New assets

1142 Question: Are new assets introduced in the paper well documented and is the documentation
1143 provided alongside the assets?

1144 Answer: [Yes]

1145 Justification: The released code is documented with instructions for data preparation,
1146 training, evaluation, and reproduction of the main tables. The release is anonymized for
1147 submission and will include license information.

1148 Guidelines:

- 1149 • The answer [N/A] means that the paper does not release new assets.
- 1150 • Researchers should communicate the details of the dataset/code/model as part of their
1151 submissions via structured templates. This includes details about training, license,
1152 limitations, etc.
- 1153 • The paper should discuss whether and how consent was obtained from people whose
1154 asset is used.
- 1155 • At submission time, remember to anonymize your assets (if applicable). You can either
1156 create an anonymized URL or include an anonymized zip file.

1157 14. Crowdsourcing and research with human subjects

1158 Question: For crowdsourcing experiments and research with human subjects, does the paper
1159 include the full text of instructions given to participants and screenshots, if applicable, as
1160 well as details about compensation (if any)?

1161 Answer: [N/A]

1162 Justification: The paper does not involve crowdsourcing or human-subject experiments, so
1163 IRB approval or equivalent review is not applicable.

1164 Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing, user studies, or research with human subjects. All experiments are conducted using publicly available datasets and automated retrieval evaluation protocols.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: The core method development and experiments do not use LLMs as an important, original, or non-standard component. Any use of writing or formatting assistance, if any, does not affect the methodology, experiments, or scientific claims.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.